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DYNAMICS OF PERTURBED ELLIPTICAL BILLIARD TABLES

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Abstract. Dynamical billiards consist of a particle on a two-dimensional table, bouncing elastically off the boundary. The state of the system is given by two numbers: one describing the location along the curve where the bounce occurs, and another describing the incoming angle of the trajectory before the bounce. Tracking these numbers over successive bounces defines a two-dimensional area preserving map, and iterating this map gives a dynamical system first studied by Birkhoff. Although there are powerful theoretical results showing that generic (strictly convex) billiards exhibit chaotic dynamics, it is nevertheless difficult (if not impossible) to decide when a given billiard table is generic and one will often resort to numerical computations. In this paper, we employ the parameterization method to compute high order Taylor expansions of the local stable/unstable manifolds attached to periodic orbits of billiard maps. Globalizing appropriate fundamental domains locates transverse intersections, providing insight into the existence and location of chaotic invariant sets. A key step in implementing the parameterization method is computing the composition of a given polynomial with the (implicitly defined) billiard map, and we show that this can be done efficiently using the discrete Fourier transform (DFT). The DFT requires evaluating the composition on a disk in the complex plane, so that we must first extend the billiard table/map to a complex domain. This problem is addressed from a computational perspective.

Keywords: Dynamical Billiards, Chaotic Dynamics, Stable/Unstable Manifolds, Parameterization Method, Birkhoff Conjecture
2020 AMS Subject Classifications: 37C83, 37M21, 37M05, 37D10

1. Introduction

The game of billiards, beloved since the 14th century, inspired the mathematical models proposed by Birkhoff in his 1927 paper [Bir27]. A goal of this paper was to solidify the foundations of thermodynamics by restricting attention to a single gas particle. Polygonal billiards on rational tables are intimately related to geodesic flows on surfaces [Gut86], and generalizations of Birkhoff's models have applications in classical mechanics and optics [BFB+11, BDBT23, DHN03, BK96]. The reader will find more complete discussion of this history, as well as many additional references, in the excellent sources of [Gut86, Tab09, CM06, HW13, BFB+11, Bun19, BFG+22, Fra03, Gut12, LT07, MRRTS16, Mei92].

Mathematical billiards are introduced formally in Section 2.1 but, loosely speaking, the idea is to consider an infinitesimal particle moving without friction on a planar billiard table. The particle bounces off the table's boundary in an elastic collision and moves off in a new direction determined by the law of specular reflection. That is, the angle of reflection, measured with

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respect to the tangent to the boundary, is equal to the angle of incidence. If the boundary of the table is a simple closed curve, then location along the boundary is described by a single number. The angle of incidence is a second number and, taken together, these describe the state of the system at the moment a bounce occurs. In between collisions, the particle moves in a straight line with constant velocity, and tracking this pair of numbers at consecutive bounces reduces the system from a differential equation with impacts to a smooth map on a cylinder. The reduced system is known as a billiard map. When the boundary is strictly convex, the billiard map is an area preserving twist map. If the boundary is at least C^3, then orbits exist for all time (i.e. the flow is complete). The relation between the flow on the table and the billiard map is illustrated graphically in Figure 1 for a pair of tables which, while simple, already suggest that billiard dynamics can be quite complicated.

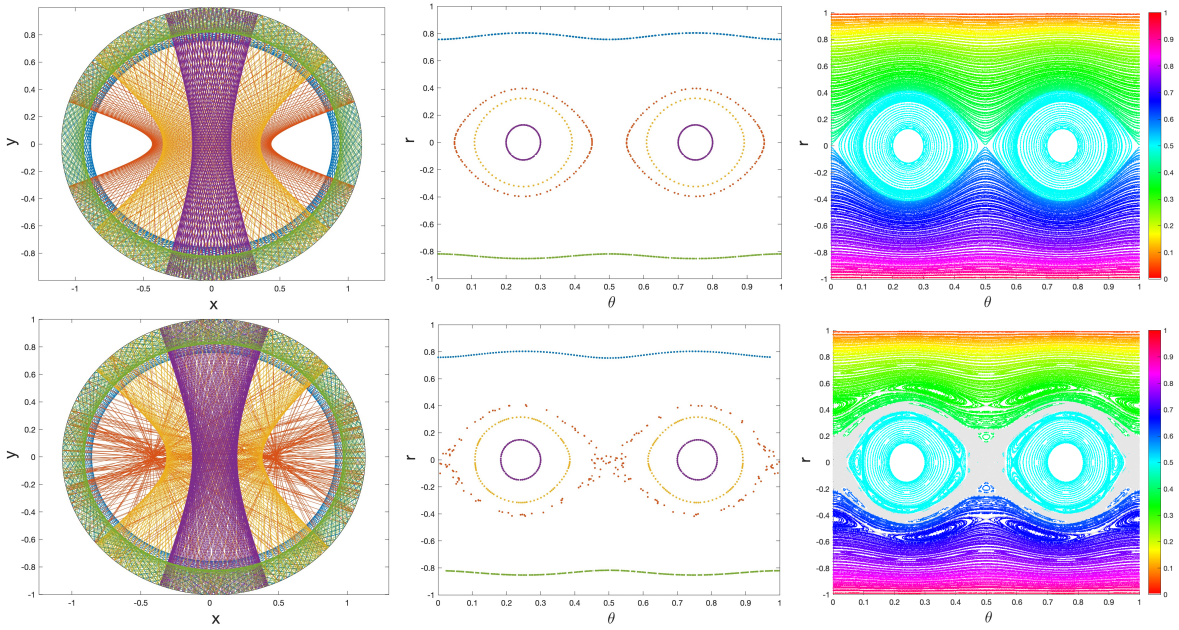


FIGURE 1. Top Left: Orbits on an elliptic table with eccentricity ≈ 0.4166. Each trajectory has a caustic, i.e. a curve which is tangent to every line. Top Middle: Phase space orbit corresponding to the trajectory on the top left. The value of θ indicates the location of bounce, and r measures the angle of the bounce. Top Right: The full phase space of orbits is quasiperiodic, colored by their rotation number. Bottom Left, Middle: The table trajectories and corresponding phase space orbits, but for a small perturbation of the ellipse (Table A described in (2) and Table 1). Note that some caustics remain, but that the red trajectory no longer has a caustic; its corresponding orbit is chaotic in phase space. Bottom Right: For the full phase space, many quasiperiodic orbits persist and are colored by frequency, but some orbits are now chaotic, colored in gray.

The discussion above makes it clear that the dynamics of a billiard map are completely determined by the shape of the boundary of the table. An important conjecture in this context

is the Birkhoff conjecture, which claims that a strictly convex billiard map is integrable if and only if the table is an ellipse (this includes the case of the circle). The Birkhoff conjecture serves as an organizing center for the theory and we review the literature in Section 2.2. See also [KS18a, BFG+22, Sch22]. For the sake of the present introductory discussion we simply remark that, while elliptic billiards are completely understood, the intuition informed by the Birkhoff conjecture is that non-elliptic tables result in richer dynamics.

Indeed, the Birkhoff conjecture is closely related to questions about chaotic dynamics. Chaos in billiards was first established by Bunimovich in the 1970s [Bun75, Bun79]. He explicitly constructed tables with only focusing components (focusing trajectories like a lens) and proved that they admit chaotic dynamics. However, the example of Bunimovich is not strictly convex (his "stadium" like table has flat boundary segments).

There is by now a well developed literature for strictly convex billiards showing that, under various generic assumptions, typical billiard maps are chaotic. This literature is reviewed in Section 2.3, but it must be stressed that care is needed when appealing to genericity results. Billiard maps are not generic in the category of area preserving twist maps, nor are real analytic billiard tables generic in the C^k category (even when $k = \infty$). When we fix our attention on a particular example, the simple fact that the boundary of the table can be written down means it is unlikely to be generic (though there must be generic tables nearby).

Because of this, numerical simulation is still among the best ways to investigate the properties of explicit examples, and in this paper we develop computational tools for studying chaotic dynamics in billiards. More precisely, we implement a flexible and accurate numerical scheme based on the parameterization method of Fontich, Cabré, and de la Llave [CFdlL03a, CFdlL03b, CFdlL05] for computing high order Taylor expansions of local stable/unstable manifolds attached to periodic orbits in billiard maps. Iterating appropriate fundamental domains allows us to "grow" the local manifolds, often locating transverse homoclinic intersections.

Assuming that the billiard table has a real analytic boundary, our series expansions are guaranteed to converge, and heuristic a-posteriori indicators provide practical guides for determining reliable domains for the truncated series. While the results of the present work are strictly numerical, we note that the parametrization method is well suited for computer-assisted proofs. See for example [MJM13, CGMMJ22]. In a follow-up paper the computations described here will be validated, leading to computer assisted existence proofs for chaotic dynamics in explicit billiard problems.

To illustrate the results obtained with our method, Figure 2 depicts stable and unstable manifolds, along with some transverse homoclinic intersections, for a variety of periodic orbits for the table of Figure 1. Figures 3 and 4 show stable and unstable manifolds with transverse homoclinic intersections for four other perturbed elliptical tables.

The desired series expansions are computed via an iterative procedure which requires computing the composition (on the level of power series) of the billiard map with the current polynomial approximation. If the billiard map were polynomial, such a composition could be worked out explicitly via Cauchy products, resulting in explicit recursion relations for the power series coefficients. See, for example, the worked problems in [GMJ17, FJ20], and also Appendix B of the current work.

Elementary non-polynomial nonlinearities can be treated by appending the differential equations describing the nonlinear terms, and several examples of this procedure are given

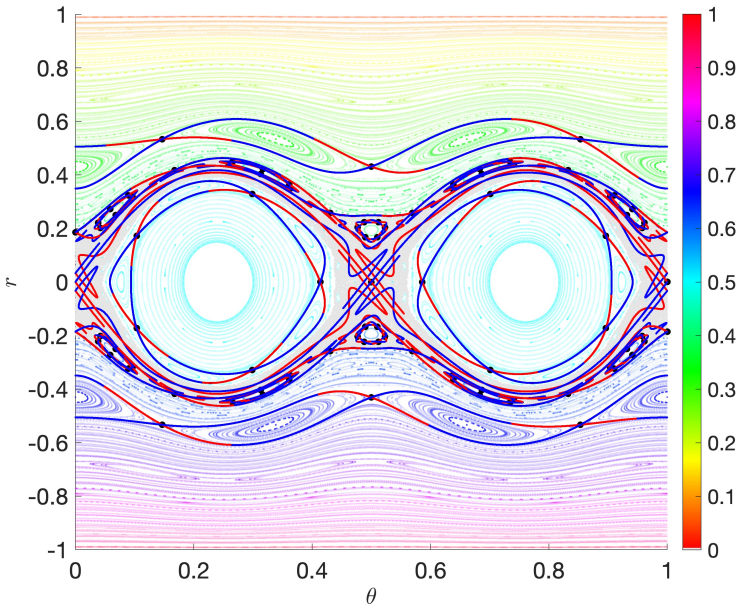


FIGURE 2. This figure shows the stable and unstable manifolds and their intersections for periodic orbits with a variety of periods. This data is for the same perturbed elliptical billiard table as shown in Fig. 1, (Table A in Tbl. 1).

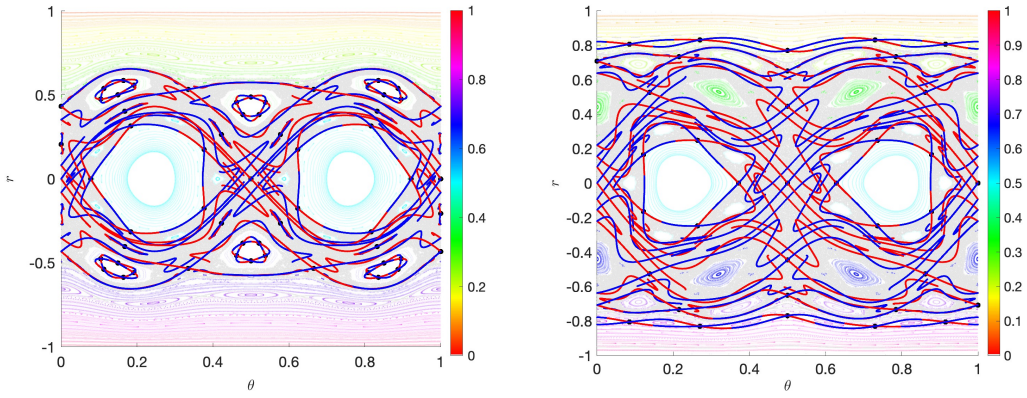


FIGURE 3. The figure shows stable and unstable manifolds for periodic orbits computed for Tables B and C (cf. Tbl. 1), larger perturbations of the same ellipse as in Figs. 1 and 2. The phase space in the background is colored by frequency for regular orbits and gray for chaotic orbits.

in the first reference just cited, and also in Appendix B. Moreover, formal series techniques for implicitly defined maps are developed in [dlLL12, TMJ22].

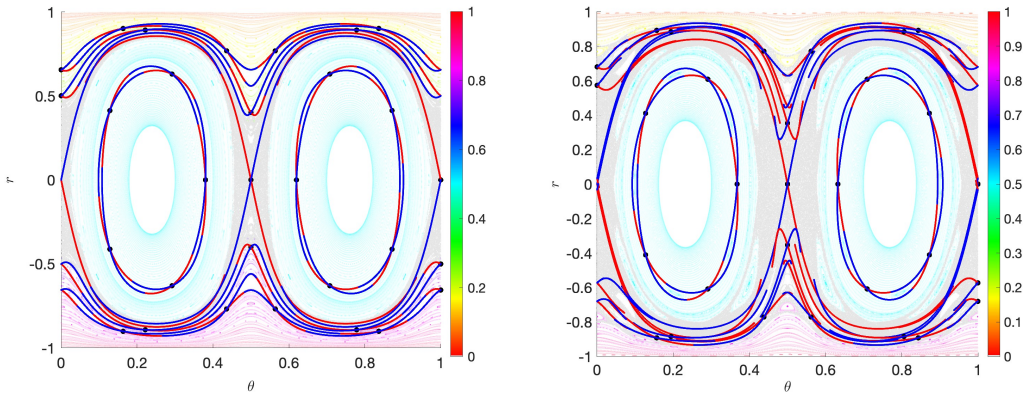


FIGURE 4. The figure shows stable and unstable manifolds for periodic orbits computed for Tables D and E (cf. Tbl. 1). These tables are perturbations of an ellipse with larger eccentricity. The phase space in the background is colored by frequency for regular orbits and gray for chaotic orbits.

Nevertheless, when the implicit map is complicated enough, the techniques in the references just cited are unwieldy. Examples of such situations include Poincaré maps associated with differential equations, and also the billiard maps studied in the present work. In these situations, an alternative is to compute the desired composition via interpolation. The discrete Fourier transform (DFT) is known to provide especially accurate, stable, and efficient interpolation for power series, and is employed throughout our computations.

A complication is that the DFT requires evaluating the composition on the boundary of a disk in the complex plane. This involves non-trivial modifications of the numerical implementation of the map. We refer to the recent work of Kumar [KAdlL25, KAdlL22, KAdlL21], where the parameterization method is implemented for Poincaré sections for problems in Celestial Mechanics applications. There, it was necessary to extend both the numerical integrator, and the surface of section in an analytic way.

In the present work, we perform a similar complex extension for billiards. That is, we extend the boundary curve to $\mathbb{C}^2$, and define a complex billiard map by considering particle motion on a complex line. We make no claim that this complex billiard map has physical significance. Rather, it is a numerical convenience. The manifolds we compute are ultimately real. This said, it is interesting to note that analytic continuation for billiards has appeared in the theory, and we refer for example to the discussion following Remark 9 in the introduction of the paper [KS18b], where the authors explain that the main idea of their proof is to analyze the singularities in the complex plane associated with the action angle form of perturbed elliptic billiards. Note that the coordinates used in the reference just cited are chosen for their theoretical properties, and do not appear to be easy to work with numerically.

The coordinates used in the present work, on the other hand, are chosen to obtain reliable numerical calculations. Indeed, for the purposes of the present work, the most important thing is to have a single formulation of the complex map which works uniformly well throughout the complex phase space. Because of this, we avoid certain simplifications which could be

used to improve efficiency of the map evaluation except on sets of measure zero. Our work builds on results in the PhD Thesis of the first author [Bis24].

We note that the focus on (strictly) convex boundaries in the present work is mainly for convenience. Non-convex boundaries lead to discontinuities, making numerical iteration more delicate. This said, our method applies in principle to periodic orbits for non-convex billiards, so long as the boundary of the table is analytic in a neighborhood of the periodic orbit. That is, as long as the table is piecewise real analytic, and the periodic orbit does not "collide" with any "joint". We will return to this topic in a future work.

The remainder of the paper is structured as follows: Section 2 introduces the setup, definitions, notation, and prior results. Section 3 gives the details of our numerical methods for iterating the billiard map on any perturbed elliptical table with smooth boundaries that is convex. In Section 4, we discuss how we compute the stable and unstable manifolds for periodic orbits. In Section 5, we detail our results, presenting computed stable and unstable manifolds for periodic orbits on five different billiard tables shown in Figures 9–17. Section 6 contains our conclusions and an outline of future work.

2. Definitions, background, and prior results

This section introduces the definitions and notation along with a review of prior results.

2.1. Dynamical billiards – from physical table to phase space.

We focus on connected, strictly-convex, planar domains with smooth (in fact real analytic) boundaries, as done in [HW13]. Such a billiard system is given by the following data.

Definition 2.1. *A billiard system consists of a point particle confined to move in a connected, strictly convex, and closed planar domain $\mathcal{D} \subset \mathbb{R}^2$ with boundary curve Γ. We assume that Γ is a smooth non-self-intersecting curve. We further assume that Γ is sufficiently smooth that it can be parameterized by a real analytic map $B : [0,1] \to \mathbb{R}^2$, where $B(0) = B(1)$, and $B'(\theta)$ is a nonvanishing tangent vector. We assume that B has counter-clockwise orientation on Γ.*

The point particle moves within domain $\mathcal{D}$ according to the following criteria.

1. *The trajectory curve $G(t) \in \mathcal{D}$ for all $t \geq 0$. We require $G(t)$ to be a continuous union of line segments, such that for any time interval for which G is in the interior of $\mathcal{D}$, $\mathbf{v}(t) = G'(t)$ is constant, and $|\mathbf{v}(t)| = 1$.*
2. *The trajectory of the particle begins at $t = 0$ at a point $G(0) \in \Gamma$ on the boundary of the domain.*
3. *Whenever the trajectory $G(t)$ reaches the boundary, it is reflected elastically back into the interior of $\mathcal{D}$ according to the law of specular reflection.*

Based on the above definition, the particle trajectory is fully determined by the location of the boundary collisions, along with the direction of vector $\mathbf{v}$ after the collision. Let $\{t_k\}_{k=0}^{\infty}$ be a discrete increasing sequence of times at which the particle hits the boundary Γ, and assume that the particle experiences no collisions for $t_k < t < t_{k+1}$. Define θ_k by $G(t_k) = B(\theta_k)$. Let γ_k be the angle that $\mathbf{v}(t)$ makes with the tangent vector $B'(\theta_k)$ immediately after the bounce, and let

$$r = \cos\gamma, \text{ where } -1 < r < 1.$$

Then the vector (θ_k, r_k) defines the k-th point of collision[1]. The restriction on r is due to the fact that $|r| \geq 1$ corresponds to the particle leaving the convex billiard table.

For a given boundary parametrization $B(\cdot)$, there is a uniquely determined diffeomorphism f on $S^1 \times (-1, 1)$, which is as smooth as the boundary, such that

$$f \begin{pmatrix} \theta_k \\ r_k \end{pmatrix} = \begin{pmatrix} \theta_{k+1} \\ r_{k+1} \end{pmatrix}.$$

When the boundary is strictly convex, it is standard to extend continuously to $r \in [-1, 1]$ by taking $f(\theta, \pm 1)$ fixed.

Birkhoff showed in [Bir27] that, if Γ is a convex curve, then f is an area preserving twist map. See also [HW13, Fra03].

Before moving to the general case, we review the simplest smooth tables.

Lemma 2.1 (Circular Table)**.** *The billiard map on a circular table with boundary parameterized by*

$$B(\theta) = \begin{pmatrix} \cos(2\pi\theta) \\ \sin(2\pi\theta) \end{pmatrix}, \ 0 \leq \theta < 1$$

has the following explicit solution:

$$f \begin{pmatrix} \theta \\ r \end{pmatrix} = \begin{pmatrix} \theta + C(r) \\ r \end{pmatrix},$$

where $C(r) = \arccos(r)/\pi$ depends only on r. In particular, an orbit starting at (θ_0, r_0) satisfies $r_k = r_0$ for all k, and $\theta_k = \theta_0 + kC(r_0) \pmod 1$.

This implies that for the circle, orbits consist of rigid rotation on horizontal lines in the phase space θ versus r. The proof of this result is due to Birkhoff [Bir27].

Lemma 2.2 (Ellipse)**.** *For any ellipse, the function f is fully integrable, meaning that there is a continuous non-constant quantity (integral) that is preserved along each trajectory.*

Interestingly, for an elliptical table, the eccentricity of the caustic conic section of each physical orbit can be used as an integral for orbits of f in the phase space. Also of note for an ellipse, all orbits are quasiperiodic, periodic, or homoclinic to a period-two orbit. See [Sib04] p.42–43 for more details. We will give more details on how to compute f in the next section.

2.2. **Birkhoff conjecture.** As already remarked above, when the boundary of a table is strictly convex, then its billiard map fixes top and bottom boundary circles of the cylinder (the boundary circles are invariant with rotation number zero). A celebrated result of Lazutkin says that, for these strictly convex billiard maps, there exists a full measure Cantor set of invariant circles in a neighborhood of the fixed boundary circles [Laz73]. Their rotation numbers have positive measure and accumulate to zero. Moreover, these Lazutkin curves are robust under perturbations by a result of Popov [Pop94].

The concept of integrability asks that the set of invariant circles be even richer. More precisely, one says that the area preserving twist map f on the plane is *locally integrable* if an open subset of its phase space is foliated by smooth invariant curves (tori), and says that f is *globally integrable* if this is true on an open and dense subset of the phase space. This

[1]To avoid cumbersome notation, when we talk about a vector, unless otherwise specified, we implicitly assume that it is a column vector.

definition makes room for the fact that a globally integrable map will often have separatrices of measure zero where the dynamics is not described by invariant tori. Note that we are maintaining a certain informality, in that we do not define caustics for billiards, which allow one to re-frame the discussion on the table rather than in the phase space of the map. The interested reader is directed to the review of [KS18a] for more precise definitions.

This said, we paraphrase the Birkhoff conjecture as follows: a strictly convex billiard map is globally integrable if and only if its table is an ellipse. The conjecture is first articulated in a paper of Poritsky in 1950 [Por50]. However Poritsky was a postdoc at Harvard in the late 1920's working with Birkhoff, and he attributes the question to his supervisor. The hypothesis of strict convexity is essential, as a theorem of Mather shows that vanishing curvature at a single point destroys the Lazutkin curves, and with them all hope of integrability.

A first major step in the direction of a resolution is found in the work of Bialy, who showed that, if the phase space of a billiard map is completely foliated by invariant circles which are not null-homotopic, then the table is a circle [Bia93]. Note that this is a stronger assumption than global integrability, as no set of measure zero is excluded in the hypothesis. Later, the algebraic version of the conjecture was established by Glutsyuk [Glu21]. This says that, if the integral of motion is polynomial in the velocity variables, then the table is an ellipse.

In [DRR96], Delshams and Ramírez-Ros proved the first perturbative version of the Birkhoff theorem, showing that symmetric, non-constant, entire perturbations of the ellipse destroy integrability. That is, they showed that – within a certain category of perturbations – an integrable billiard close to an ellipse must be an ellipse. This result was extended to more general perturbations of the circle by Avila, De Simoi, and Kaloshin in [ADSK16]. Another deep result in this direction was established by Kaloshin and Sorrentino in [KS18b], which shows that if a billiard map has enough rational invariant circles, then it is an ellipse. In Appendix F of the paper just cited, the authors sketch a proof of the complete global (non-perturbative) Birkhoff conjecture, introducing a novel curve shortening flow. They are however careful to say that completing this sketch would require overcoming serious technical obstacles.

Very recently an extension by Koval has appeared showing that, for almost every eccentricity e, a smooth enough integrable billiard sufficiently close to an e-ellipse is itself an ellipse [Kov26]. Note that the notion of integrability considered here focuses on the dynamics near the fixed boundary circles, and that this is one of the most complete results to date: establishing a version of the Birkhoff conjecture except on a set of measure zero. Another recent result due to Bialy and Mironov shows that centrally-symmetric billiard tables satisfy the Birkhoff conjecture [BM22], and even more recently Kaloshin, Koudjinan, and Zhang have completed a perturbative extension [KKZ24].

Note that an interesting and subtle point has emerged in recent years. Namely, Treschev has constructed a numerical example of a system which appears to be locally integrable in an open neighborhood of its period-two orbit, far from the fixed boundary circles of the billiard map, without being locally integrable near the boundary circles [Tre13]. If this turns out to be true, it would show that the "inner" and "outer" dynamics of billiard maps (far from and near the boundary respectively) are more independent than once suspected. In this case, the Birkhoff conjecture would be rephrased as follows: every billiard which is locally integrable near its boundary is an ellipse. This would not lead to any contradiction, as a close look at the notions of integrability in all of the preceding works shows that the hypotheses treat a neighborhood of the boundary.

Finally we note that the higher dimensional situation seems less clear. While billiards with ellipsoid boundaries are integrable, Mironov and Yin have shown that certain conical tables in $\mathbb{R}^d$ also lead to integrable billiards [MY26]. The appropriate generalization of the Birkhoff conjecture in $\mathbb{R}^d$ has, to the best of our knowledge, not yet emerged.

2.3. **Chaotic dynamics.** A complementary line of research asks: how common is chaotic behavior in billiard systems? Let us focus once again on smooth convex boundary curves. A theorem of Donnay in this context shows that there exist C^∞ curves, C^2 close to the ellipse, having periodic orbits which generate homoclinic tangles [Don03]. These systems therefore have positive topological entropy and are not integrable.

A powerful C^2 result of Carneiro, Kamphorst, and Pinto-de-Carvalho [DCOKPdC07], proven independently in the C^∞ category by Xia and Zhang [XZ14], says that generic billiards have (i) finitely many orbits of period K, (ii) at least one of these period K orbits is hyperbolic, and (iii) the stable and unstable manifolds of the hyperbolic orbit intersect transversally. Again, this gives rise to invariant sets with positive topological entropy and rules out integrability. A very recent work by Baldomá, Florio, Leguil, and Seara extends this result to billiards with real analytic boundaries [BFLS26]. These results show that chaos is "typical" in billiard maps. Generalizations to higher dimensional billiards are found for example in the works of Delshams, Fedorov, and Ramírez-Ros [DFRR01] as well as in the work of Bessa, Del Magno, Lopes Dias, Gaivão, and Torres [BDMLD+24].

Many other studies have focused on numerical simulation of billiard systems, and we refer the interested reader to Levi and Tabachnikov [LT07] for a method which utilizes the minimization property to find iterates of the map. We also note that Lansel and Porter [LP04] and [Tur16] simulate billiard systems that are not necessarily convex, while Solanpää et al. [SLR16] consider more general nonconvex billiard systems with multiple charged particles subject to magnetic fields. In [dCHSL22], da Costa et al. describe billiard dynamics on oval-like tables using a polar equation, with a circle approximation to estimate the next point of contact with the boundary. See also [AV21, BFB+11, CF21, Dat17, Kni98, MRRTS16, PRK22, PTZ23, RGK21]. We only note that none of these works required numerical continuation of the billiard map to complex tables and lines.

In [RR05] Ramírez-Ros numerically computes high order expansions for the stable/unstable manifolds of the period two orbit in some perturbed elliptic billiards, in order to study exponentially small separatrix splitting for these systems. The philosophy behind these computations is quite different from the one developed in the present work, as the author expands around the known local stable/unstable manifolds of the period two point in the elliptic billiard. In particular, the author of the work just cited computes coefficients by exact recursion rather than by interpolating via the DFT. On the other hand, our computations are designed to work for arbitrary periodic orbits and far from the perturbative regime.

3. Numerical computation of the billiard map

We now discuss our numerical procedure for evaluating the real billiard map.

3.1. **Table shapes.** Note that an ellipse with eccentricity $\sqrt{1 - b_1^2/a_1^2}$ has parameterization

$$E(\theta) = \begin{pmatrix} a_1 \cos(2\pi\theta) \\ b_1 \sin(2\pi\theta) \end{pmatrix}. \tag{1}$$

We focus on perturbed ellipses of the form $B(\theta) = (x(\theta), y(\theta))$ with Fourier series given by

$$
\begin{aligned}
x(\theta) &= a_{1,1}\cos(2\pi\theta) + \sum_{k=2}^{n} a_{k,1}\cos(2k\pi\theta) + b_{k,1}\sin(2k\pi\theta) \\
y(\theta) &= b_{1,2}\sin(2\pi\theta) + \sum_{k=2}^{n} a_{k,2}\sin(2k\pi\theta) + b_{k,2}\sin(2k\pi\theta).
\end{aligned} \tag{2}
$$

Table	A	B	C	D	E
$(a_{1,1}, a_{2,1}, a_{3,1})$	(1.1,0.03,0)	(1.1,0.05, 0.00015)	(1.1, 0.08, 0.0002)	(2, 0.04,0)	(2, 0.05,0)
$(b_{1,2}, b_{2,2}, b_{3,2})$	(1,0.03,0)	(1,0.035, 0.0001)	(1, 0.095, 0.0001)	(1, 0.035,0)	(1, 0.065,0)
Eccentricity	0.4166	0.4166	0.4166	0.8660	0.8660

TABLE 1. The coefficients used for our choice of perturbed elliptical billiard tables, referred to as Billiard Tables A–E. Any unspecified coefficients are zero. The bottom row states the approximate eccentricity of the associated unperturbed ellipse.

Notice that for a table specified by (2), if $a_{1,1}$ and $b_{1,2}$ are the only nonzero coefficients, then the table is an ellipse. In particular, for any table, the *ellipse associated with this table* is the ellipse with the same $a_{1,1}$ and $b_{1,2}$ values (all other coefficients set to zero). The particular tables we have used for our numerics are referred to as Billiard Tables A–E. The coefficients $a_{k,i}$ and $b_{k,i}$ are given in Tbl. 1. For Tables A-C, the associated ellipse has $a_{1,1} = 1.1$ and $b_{1,2} = 1$. For Tables D and E the associated ellipse has $a_{1,1} = 2$ and $b_{1,2} = 1$. Tables A-E and the associated ellipses are depicted in Figure 5.

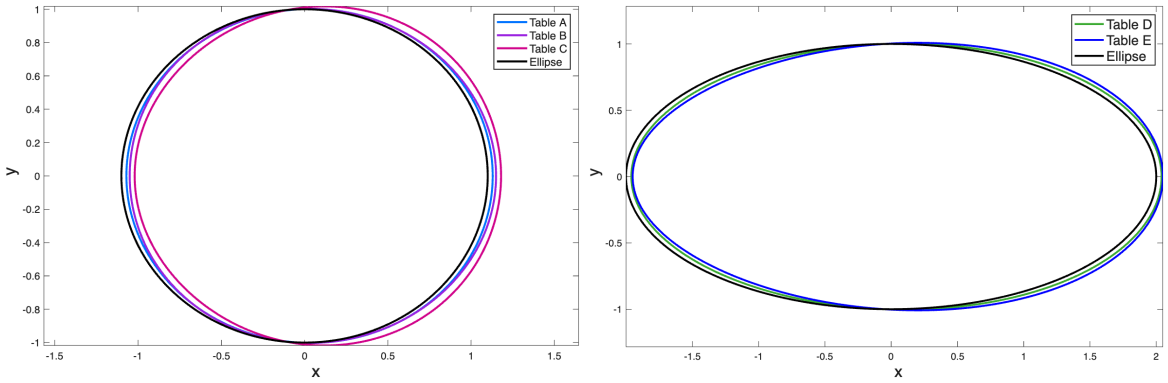


FIGURE 5. Shape of billiard tables used in the numerics. (Left) Billiard Tables A-C are perturbations of an ellipse with eccentricity ≈ 0.4166. (Right) Billiard Tables D and E are perturbations of an ellipse with eccentricity ≈ 0.8660.

Since ellipses are convex and since $B(\theta)$ varies continuously with respect to the coefficients, the table remains convex when the higher-order Fourier coefficients are sufficiently small – specifically, if for $i = 1, 2$ and for all $k > 1$, $|a_{k,i}|$ and $|b_{k,i}|$ are significantly smaller than $|a_{1,1}|$

and $|b_{1,2}|$, the table remains convex. To illustrate how convexity is lost as these perturbations grow, consider the parameterized table with coefficients given by:

$$(a_{1,1}, a_{2,1}) = (1.1, 0.03\epsilon)$$
$$(b_{1,2}, b_{2,2}) = (1, 0.025\epsilon). \tag{3}$$

For $\epsilon = 0$, this table is an ellipse. Figure 6 (left panel) shows how the table shape changes as ϵ increases, while the right panel shows the minimum value of the signed curvature measured along the boundary of the table. At $\epsilon \approx 9$, the minimum signed curvature passes through zero, and the table loses convexity. The tables given by (3) have the same associated ellipse as Billiard Tables A, B, and C; in particular Billiard Tables A-C have coefficients $a_{2,1}$ and $b_{2,2}$ with $\epsilon < 4$, which is clearly before convexity is lost.

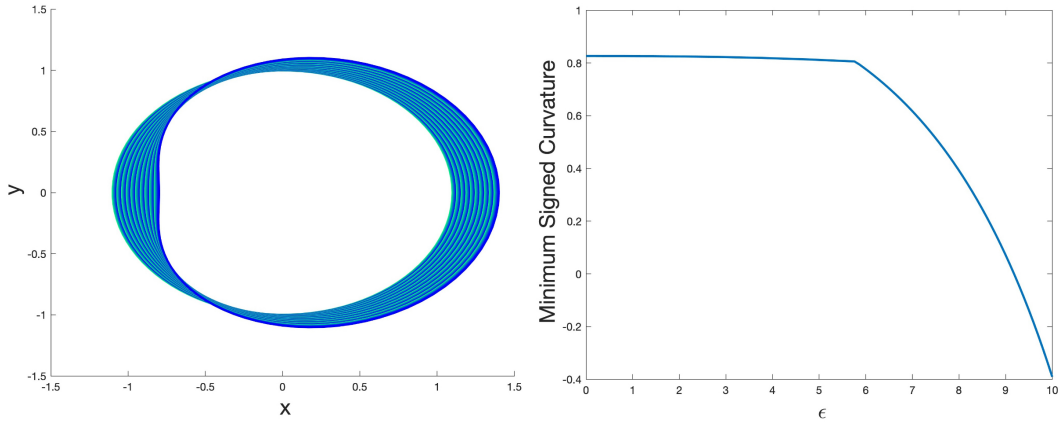


FIGURE 6. (Left) The shape of a table with the nonzero coefficients given by (3) for values of ϵ between 0 and 10. (Right) The minimum signed curvature of the table. When $\epsilon \approx 9$, the signed curvature is zero at a point on the boundary, implying that the table has lost convexity.

3.2. **Iteration for elliptical tables.** In order to compute the billiard map for general tables, we use the fact that we can directly compute the billiard map for an ellipse. We compute the location of each bounce on the table's corresponding ellipse, then use this as the initial guess for Newton's method to find the location of the bounce on the perturbed table. In this section we explain how to compute the ellipse case.

Let $B(\theta) = (x(\theta), y(\theta)) = (a_1 \cos(2\pi\theta), b_1 \sin(2\pi\theta))$ parameterize an ellipse, as introduced in (1). Given an input (θ, r), we seek $(\hat{\theta}, \hat{r})$ such that $f(\theta, r) = (\hat{\theta}, \hat{r})$. Based on r, we can use the arccosine function to find the direction vector $\mathbf{v}$. Since $B(\theta)$ and $B(\hat{\theta})$ must lie on the line parallel to the vector $\mathbf{v} = (v_1, v_2)$, there exists s_* such that

$$B(\hat{\theta}) = B(\theta) + s_*\mathbf{v}. \tag{4}$$

Let

$$d_1 = v_1/a_1 \text{ and } d_2 = v_2/b_1.$$

Combining these equations with the equation for an ellipse and solving for s_* gives

$$s_* = -2\left(\frac{d_1\cos(2\pi\theta) + d_2\sin(2\pi\theta)}{d_1^2 + d_2^2}\right). \tag{5}$$

Plugging this value back into (4) to get $B(\hat{\theta}) = (\hat{x}, \hat{y})$. We find $\hat{\theta}$ from $B(\hat{\theta})$ since

$$\frac{b_1}{a_1}\tan(2\pi\hat{\theta}) = \frac{\hat{y}}{\hat{x}}.$$

This last statement is where the simplification of working with the ellipse is really clear, since for a perturbed ellipse, we cannot solve directly and must instead use a two-dimensional root finding method. Finally, from this point finding $\hat{r}$ is exactly the same as for the perturbed case, and it will be explained in the next section.

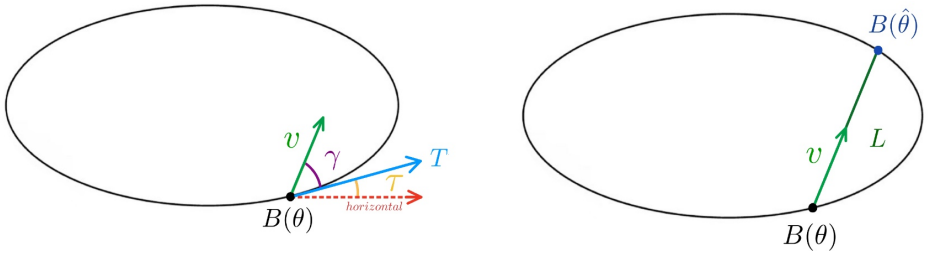


FIGURE 7. Notation for our numerical methods. Left: $\mathbf{v}, \gamma, T$, and τ. Right: The line L contains $B(\theta)$ and $B(\hat{\theta})$.

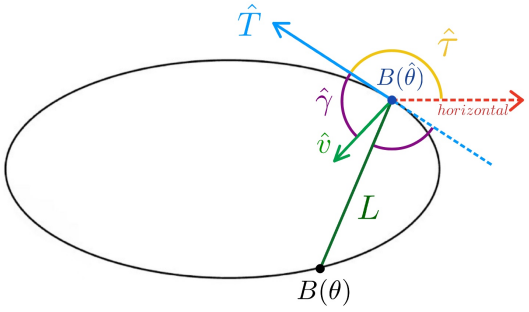


FIGURE 8. Computation of the $\hat{r}$ from $\mathbf{v}$ and $\hat{\theta}$.

3.3. **Iteration for general billiard tables.** Given θ and r, we use the following notation, depicted in Fig. 7 (left).

(1) $T = B'(\theta)$ the tangent vector to the boundary curve.

(2) γ is defined implicitly by $r = \cos\gamma$. We therefore define it directly by $\gamma = \arccos r$.

(3) τ is the angle between the positive horizontal direction and T. Thus τ is defined implicitly by $T_2/T_1 = \tan\tau$. To find τ, we use the quadrant-specific version of the arctan function (`atan2` in MATLAB) and set $\tau = \texttt{atan2}(T)$.

(4) $\mathbf{v}$ is defined implicitly by saying that γ is the angle between $\mathbf{v}$ and T. From the previous two definitions, we get

$$\mathbf{v} = \begin{pmatrix} \cos(\gamma+\tau) \\ \sin(\gamma+\tau) \end{pmatrix}.$$

Using the notation defined above, we are ready to find $\hat{\theta}$ using the fact that $B(\theta)$ and $B(\hat{\theta})$ lie on a line parallel to $\mathbf{v}$, given by

$$L(s) = B(\theta) + s\mathbf{v}, \tag{6}$$

where $s \in \mathbb{R}$ is a scalar. See right panel of Figure 7. Since the table is convex, this line has exactly two intersection points with the boundary, the first being the original point $B(\theta)$ at $s = 0$. The second point $B(\hat{\theta})$ occurs at $s > 0$. As long as we have a good guess near the desired solution, this intersection can be formulated as a zero finding problem

$$\mathbf{0} = h(s,\hat{\theta}) = L(s) - B(\hat{\theta}), \tag{7}$$

where the function $h : \mathbb{R}^2 \to \mathbb{R}^2$. Since L and B are both smooth, we solve using Newton's method. It seems counterintuitive to solve for $\hat{\theta}$ and s rather than solving only for $\hat{\theta}$. However, a one-variable root finding problem for $\hat{\theta}$ alone has higher-multiplicity zeroes in some important cases (specifically, whenever $\mathbf{v}$ is orthogonal to T).

We now explain how to find a good initial guess $(s_{\text{guess}}, \hat{\theta}_{\text{guess}})$ for our Newton's method to find a root for h. We saw in the last section that if f_e is the billiard map for an ellipse, then the iterate $f_e(\theta_e, r_e) = (\hat{\theta}_e, \hat{r}_e)$ can be computed in an explicit way, and this calculation also yields s_*. Therefore, given the billiard map f_e for the ellipse associated with our table, we compute $f_e(\theta, r) = (\hat{\theta}_e, \hat{r}_e)$. We use $s_{\text{guess}} = s_*$, the s value for the ellipse given in (5) and $\theta_{\text{guess}} = \hat{\theta}_e$ for our guess for Newton's method. This is usually a good enough guess to allow us to find $(\hat{\theta}, \hat{r})$ for the perturbed table.

Though the method above works the vast majority of the time, it is not enough to guarantee convergence to the correct root in some small number of cases. We make two modifications. The first modification takes into account that when the ball only bounces a short distance, or more specifically $|r|$ is close to 1, then Newton's method is likely to find the spurious root that occurs at $s = 0$. In order to avoid this problem, and make sure that we converge to the correct root in these cases, we use the method of *deflation* on the function h to eliminate the incorrect zero to which we do not want Newton's method to converge. That is, since we know that h has a root at $(s, \hat{\theta}) = (0, \theta)$, we modify h as follows

$$\tilde{h}(s,\hat{\theta}) = \frac{h(s,\hat{\theta})}{\|(s-0, \hat{\theta}-\theta)\|}.$$

This has the same second root as h but avoids the spurious solution. For a more complete discussion on deflation see [BG71, PW71, Bee15, FBF15].

The second modification is a technical modification in making sure that the initial guess is well defined. It corrects for the fact that $\mathbf{v}$ for the perturbed table might point outside the associated ellipse table. See the discussion and Figure 18 in Appendix A.

After we have found $\hat{\theta}$, the last step is to find $\hat{r}$. Using similar notation as before, we use the notation below. See Fig. 8.

(1) $\hat{T} = B'(\hat{\theta})$ is the tangent vector.
(2) $\hat{\tau}$ is the angle between the tangent line and the positive horizontal axis.
(3) $\hat{\rho}$ is the angle between $\mathbf{v}$ and the positive horizontal axis.
(4) $\hat{\gamma} = \hat{\rho} - \hat{\tau}$. This is the angle between $\mathbf{v}$ and $\hat{T}$, which is up to sign the same as the angle between $\hat{T}$ and the reflected vector $\hat{\mathbf{v}}$. (The value of $\hat{r}$ depends only on $\hat{\gamma}$ and not on $\hat{\mathbf{v}}$.)
(5) $\hat{r} = \cos\hat{\gamma}$.

We have found this method to work without failure for small perturbations of ellipses. In particular, since we initialize Newton's method with $(s_{\text{guess}}, \hat{\theta}_{\text{guess}}) = (s_*, \hat{\theta}_e)$, the boundary of a too greatly perturbed table may be insufficiently close to the guess to get convergence. In order to find an upper bound on the perturbation which results in Newton's method converging within a reasonable number of iterations, we studied the table with coefficients given in (3). This table was previously found to have lost convexity around $\epsilon \approx 9$ (see Figure 6), but Newton's method was found to converge reliably only for $\epsilon \in [0, 5]$. Convergence of Newton's method was studied in depth for $\epsilon \in [5, 6.5]$. We calculated orbits for an initial location of (θ_0, r_0) with $\theta_0 = 0.3$ and 100 values of r_0 ranging from -0.9 to 0.9. These initial conditions were tested on the table given by (3) for each value of ϵ, where 23 values of ϵ were chosen, ranging from $\epsilon = 5$ to $\epsilon = 6.5$. We call Newton's method non-convergent if for any bounce in the first 100 iterations of the orbit, Newton's method did not converge. Any ϵ which resulted in a non-convergent Newton's method for any initial condition was said to be too large of a perturbation. We found that for $\epsilon \in [5.0, 5.778]$, the table was sufficiently close to elliptical for our method to converge, while for $\epsilon \geq 5.8434$, the table was non-convergent often enough to make the method impractical without further modifications. This limitation is not an essential barrier for the method. Instead of starting at an initial guess at an unperturbed ellipse, we could repeat the process of guesses with a series of progressively larger perturbations until we get to a desired large perturbation.

3.4. **Continuation – analytic billiard maps.** We extend the billiard map analytically to a complex domain via the following steps.

(1) Implicitly define $\mathbf{v}$ as a function of (θ, r): We rewrite our statements about $\tau, \gamma, \theta, T, r$, and $\mathbf{v}$ implicitly as a root finding problem.

$$g(\tau, \gamma, \theta, r) = \begin{pmatrix} \cos\gamma - r \\ \\ T_1 \sin\tau - T_2 \cos\tau \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} \cos(\gamma + \tau) \\ \sin(\gamma + \tau) \end{pmatrix}$$

Note that g is perfectly well defined for complex values of τ, γ, θ, and r. Indeed, g is complex analytic in these variables.

By implicit differentiation

$$\frac{\partial \mathbf{v}}{\partial(\theta, r)} = \frac{\partial \mathbf{v}}{\partial(\gamma, \tau)} \frac{\partial(\gamma, \tau)}{\partial(\theta, r)}$$

so that

$$\frac{\partial(\gamma, \tau)}{\partial(\theta, r)} = -\left(\frac{\partial g}{\partial(\gamma, \tau)}\right)^{-1} \frac{\partial g}{\partial(\theta, r)}$$

$$\frac{\partial \mathbf{v}}{\partial \gamma} = \frac{\partial \mathbf{v}}{\partial \tau} = \begin{pmatrix} -\sin(\gamma+\tau) \\ \cos(\gamma+\tau) \end{pmatrix}$$

$$\frac{\partial g}{\partial(\theta, r)} = \begin{pmatrix} 0 & -1 \\ \dfrac{\partial T_1}{\partial \theta}\sin\tau - \dfrac{\partial T_2}{\partial \theta}\cos\tau & 0 \end{pmatrix}$$

$$\frac{\partial g}{\partial(\gamma, \tau)} = \begin{pmatrix} -\sin\gamma & 0 \\ 0 & T_1\cos\tau + T_2\sin\tau \end{pmatrix}$$

The term $\partial g/\partial(\gamma,\tau)$ is non-singular under the following condition, which holds for any well defined billiard iterate.

(a) $\sin\gamma \neq 0$ meaning that $\gamma \neq 0, \pi$. These values correspond to the cases where the direction of bounce is tangent to the boundary – which is not a possibility for any convex billiard table.
(b) $T_1\cos\tau + T_2\sin\tau \neq 0$. If this was equal to zero and $g = \mathbf{0}$, then we can show that $T_1 = T_2 = 0$.

(2) Implicitly define $(s, \hat{\theta})$ as a function of (θ, r): We quickly restate the function h so that we can write down the derivative.

$$h(\theta, r, s, \hat{\theta}) = B(\theta) + s\mathbf{v} - B(\hat{\theta}) = \mathbf{0}$$

To compute the derivative, define $\hat{T} = B'(\hat{\theta})$ and $T = B'(\theta)$. Using implicit differentiation, we get

$$\frac{\partial(s,\hat{\theta})}{\partial(\theta, r)} = -\begin{pmatrix} \mathbf{v} & | & -\hat{T} \end{pmatrix}^{-1} \begin{pmatrix} T + s\dfrac{\partial \mathbf{v}}{\partial \theta} & | & s\dfrac{\partial \mathbf{v}}{\partial r} \end{pmatrix}$$

For the part in the inverse to be singular, we would need $\hat{T}_2/\hat{T}_1 = v_2/v_1$. This only would occur if $\mathbf{v}$ and $\hat{T}$ were parallel, which cannot happen for a convex billiard table.

(3) Implicitly define $\hat{r}$ as a function of (θ, r): This is formulated implicitly as follows.

$$k(\theta, r, \hat{\rho}, \hat{\tau}) = \begin{pmatrix} \hat{T}_1\sin\hat{\tau} - \hat{T}_2\cos\hat{\tau} \\ v_1\sin\hat{\rho} - v_2\cos\hat{\rho} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \text{ where } \hat{r} = \cos(\hat{\rho} - \hat{\tau}).$$

Note that this is written in terms of $\mathbf{v}$ and not $\hat{\mathbf{v}}$. Using implicit differentiation

$$\frac{\partial \hat{r}}{\partial(\theta, r)} = \frac{\partial \hat{r}}{\partial(\hat{\rho}, \hat{\tau})}\frac{\partial(\hat{\rho}, \hat{\tau})}{\partial(\theta, r)}$$

and

$$\frac{\partial(\hat{\rho}, \hat{\tau})}{\partial(\theta, r)} = -\left(\frac{\partial k}{\partial(\hat{\rho}, \hat{\tau})}\right)^{-1}\frac{\partial k}{\partial(\theta, r)}.$$

These partial derivatives are given by

$$\frac{\partial k}{\partial(\hat{\rho}, \hat{\tau})} = \begin{pmatrix} 0 & \hat{T}_1\cos\hat{\tau} + \hat{T}_2\sin\hat{\tau} \\ v_1\cos\hat{\rho} + v_2\sin\hat{\rho} & 0 \end{pmatrix},$$

and

$$\frac{\partial k}{\partial(\theta,r)} = \begin{pmatrix} \dfrac{\partial \hat{T}_1}{\partial \theta}\sin\hat{\tau} - \dfrac{\partial \hat{T}_2}{\partial \theta}\cos\hat{\tau} & \dfrac{\partial \hat{T}_1}{\partial r}\sin\hat{\tau} - \dfrac{\partial \hat{T}_2}{\partial r}\cos\hat{\tau} \\ \dfrac{\partial v_1}{\partial \theta}\sin\hat{\rho} - \dfrac{\partial v_2}{\partial \theta}\cos\hat{\rho} & \dfrac{\partial v_1}{\partial r}\sin\hat{\rho} - \dfrac{\partial v_2}{\partial r}\cos\hat{\rho} \end{pmatrix}, \text{ where } r = \cos\gamma.$$

(4) It only remains to show that $\partial k/\partial(\hat{\rho},\hat{\tau})$ is non-singular when $k = 0$. If the determinant of the Jacobian matrix is zero, then we can show that either $\mathbf{v}$ or $\hat{T}$ is the zero vector. By our assumptions on $\mathbf{v}$ and B, that cannot happen.

Unlike in the real-valued case, it is not feasible to use an ellipse as an initial guess. In fact, we can continue the ellipse into the complex plane along different paths to get different answers. Therefore to find the iterate of (θ, r) we require an initial guess $(\hat{\theta}_{\text{guess}}, \hat{r}_{\text{guess}})$, as well as guesses approximating the complex-valued angles $\tau, \gamma, \hat{\tau}, \hat{\gamma}$. These guesses come from the exact iterate values for a point (θ_p, r_p) which is close to (θ, r), so the analytic continuation is performed using numerical continuation.

Note that the continuation version of the map can be continued to the case of $\hat{\theta}$ outside of $[0, 1)$, whereas the way that the real version was designed means that it will never be possible to find an iterate outside this range.

4. Parameterization Method for Hyperbolic Periodic Orbits of Planar Maps

The idea is to formulate a family of higher dimensional maps, whose fixed points correspond to periodic orbits of the billiard map. In addition to providing a stable method for numerically computing periodic orbits, the multiple shooting maps can be analyzed to recover linear and nonlinear stability information about the periodic orbit. These ideas are reviewed in the next several sections.

4.1. Multiple shooting.

We focus on the special case of planar maps, as this is the case used in the present work, and this restriction simplifies parts of the discussion. The material in this section and in Section 4.2 is standard, but it is essential for the remainder of the present work. We restate a number of well known results without proof in this context. We refer to [SN10, GMJ17, HLMJ23] for more complete discussion, including many additional references.

Let $S \subset \mathbb{R}^2$ be an open and connected set and suppose that $f : S \to \mathbb{R}^2$ is a smooth map. For $k \in \mathbb{N}$, let f^k denote the composition of f with itself k times, and let $f^0 = \mathrm{Id}_{\mathbb{R}^2}$ denote the identity map. We say that $\hat{u} \in S$ is a period-K point for f if

$$f^K(\hat{u}) = \hat{u}.$$

We say that $\hat{u}$ has least period K if $f^k(\hat{u}) \neq \hat{u}$ for $0 < k < K$. Letting $u_1 = \hat{u}$, and $u_j = f(u_{j-1})$ for $1 < j \leq K$, we say that the ordered collection of points $\{u_1, \ldots, u_K\}$, each in S, is a periodic orbit for f, noting that $f(u_K) = u_1$.

Fix K, let $S^K \subset \mathbb{R}^{2K}$ denote the K-fold cartesian product of S. Define the multiple shooting map $F_K : S^K \to \mathbb{R}^{2K}$ by

$$F_K(u_1, u_2, \ldots, u_{K-1}, u_K) = (f(u_K), f(u_1), f(u_2), \ldots, f(u_{K-1})). \tag{8}$$

Note that $\mathbf{u} = (u_1, \ldots, u_K) \in S^K$ satisfies $F_K(\mathbf{u}) = \mathbf{u}$, if and only if $\{u_1, \ldots, u_K\}$ is a period-K orbit of f. That is, fixed points of F_K correspond to period-K orbits of f. Moreover, K is

the least period if and only if $u_1 \neq \ldots \neq u_K$. We suppress the subscript and write $F_K = F$ when K is clear from context.

The Jacobian matrix of F is the following $2K \times 2K$ matrix

$$\mathrm{DF}(\mathbf{u}) = \begin{pmatrix} 0_{2\times 2} & 0_{2\times 2} & \cdots & 0_{2\times 2} & \mathrm{Df}(u_K) \\ \mathrm{Df}(u_1) & 0_{2\times 2} & \cdots & 0_{2\times 2} & 0_{2\times 2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0_{2\times 2} & 0_{2\times 2} & \cdots & \mathrm{Df}(u_{K-1}) & 0_{2\times 2} \end{pmatrix},$$

where $0_{2\times 2}$ denotes the 2×2 zero matrix and $\mathrm{Df}(u_i)$ is the 2×2 Jacobian of f evaluated at u_i. The virtue of the multiple-shooting framework is that it dispenses with compositions.

Fixed points of F are equivalent to zeros of the map $G : S^K \to \mathbb{R}^{2K}$ defined by

$$G(\mathbf{u}) = F(\mathbf{u}) - \mathbf{u},$$

and zeros of G are found computed using Newton's method. That is, if $\mathbf{u}^0 \in S^K$ has $\|G(\mathbf{u}^0)\|$ small enough, then the Newton sequence $\{\mathbf{u}^k\}$ defined by

$$\mathbf{u}^{k+1} = \mathbf{u}^k + \boldsymbol{\Delta}^k,$$

where $\boldsymbol{\Delta}^k \in S^K$ solves the linear equation

$$DG(\mathbf{u}^k)\boldsymbol{\Delta}^k = -G(\mathbf{u}^k),$$

converges to $\mathbf{u}_*$ with $G(\mathbf{u}_*) = 0$. The components of $\mathbf{u}_*$ provide a periodic-K orbit for f. Here we have the explicit formula

$$DG(\mathbf{u}) = DF(\mathbf{u}) - \mathrm{Id}_{2K\times 2K},$$

where $\mathrm{Id}_{2K\times 2K}$ denotes the $2K \times 2K$ identity matrix.

4.2. **Stability for periodic orbits.** The stability of a periodic orbit is defined by applying the usual notion of fixed point stability to the composition map f^K. Moreover, the eigenvalues of the matrices $Df^K(u_j)$ do not depend on $1 \leq j \leq K$. That is, each of these matrices has the same eigenvalues $\lambda_1, \lambda_2 \in \mathbb{C}$, and we refer to these as the multipliers of the periodic orbit. Note, however, that the associated eigenvectors are in general different for different j. The multipliers λ_1, λ_2 can be recovered from the Jacobian of the multiple shooting map. Indeed, by Proposition 1 of [SN10], Lemma 3.1 of [GMJ17] or Proposition 3.1 of [HLMJ23], we get the following lemma.

Lemma 4.1 (Multipliers via the multiple shooting eigenvalues)**.** *Let $\mathbf{u}_* = (u_1, \ldots, u_K)$ be a fixed point of the multiple shooting operator F associated with the map f, so that each u_j is a period-K under f.*

- *The eigenvalues of $DF(\mathbf{u}_*)$ are the K-th roots of the multipliers of the periodic orbit. More precisely, if (α, ξ) is an eigenvalue-eigenvector pair for $DF(\mathbf{u}_*)$ with $\xi = (\xi_1, \ldots, \xi_K)$, then (α^K, ξ_j) is an eigenpair for $Df^K(u_j)$, $1 \leq j \leq K$.*
- *If $\{u_1, \ldots, u_K\}$ is a period-K saddle orbit, then the multipliers $|\lambda_1| < 1 < |\lambda_2|$ are real, since the matrices $Df^K(u_j)$ are real, and each has a real eigenbasis.*
- *If (α, ξ) is an eigenpair with α and ξ complex then, since $DF(\mathbf{u}_*)$ is a real matrix, we have that $(\alpha, Real(\xi))$ and $(\alpha, imag(\xi))$ are both eigenpairs for $DF(\mathbf{u}_*)$.*

In the present work, the map f is orientation preserving, and the orbits we consider have two positive real multipliers (we did not find orbits with two negative multipliers, but we have not ruled them out). When there are two positive real multipliers, there is always a K-th root which is real, and this is the root we will always choose to work with. This leads to real eigenvectors. Note however that, more generally, if $Df^K(u_j)$ has a negative multiplier, and if K is even, then there will be no real K-th root. This, and more general cases involving higher dimensional manifolds and maps, are described in [GMJ17].

4.3. **Identifying orbit types – frequency and chaos.** If f is an area preserving map of the plane, then typical orbits of f are either periodic, quasiperiodic, or chaotic, and the method of weighted Birkhoff averages can be used to distinguish between these, as well as computing the frequency of quasiperiodic orbits. We briefly discuss this method, and refer to [DSSY17, DY18, DSSAY19, SM20, MS21] for more detailed discussion.

Consider an orbit of f $\{(\theta_k, r_k)\}_{k=0}^{\infty}$, where $(\theta_{k+1}, r_{k+1}) = f(\theta_k, r_k)$. The frequency (or rotation number) of the orbit is defined to be the average amount that the θ variable changes in a single iterate. More precisely, define

$$\text{frequency} = \lim_{N\to\infty} \frac{1}{N} \sum_{k=0}^{N-1} (\theta_{k+1} - \theta_k).$$

Note that this is a slight abuse of notation, as the frequency is actually defined on a lift of f to $\tilde{f} : \mathbb{R}\times(-1,1) \to \mathbb{R}\times(-1,1)$. Under standard smoothness and ergodicity hypotheses, the limit of this average is an invariant of the orbit and does not depend on the initial condition/initial position along the orbit. In practice however this sum converges very slowly, with complexity $O(1/N)$.

The weighted Birkhoff average is a tool which accelerates the convergence, but only along nonchaotic orbits with Diophantine frequency. This results in spectral convergence, faster than $O(1/N^k)$ for all $k \in \mathbb{N}$. The weighted Birkhoff average still converges slowly for chaotic orbits, and because of this it can be used as a sieve to distinguish chaotic from quasiperiodic orbits.

In order to distinguish chaotic orbits, given a length-N orbit segment, we compute a weighted Birkhoff average on iterates $0, \dots, N/2$, and compare the answer to the same weighted Birkhoff average for iterates $N/2, \dots, N$ (in practice it suffices to take $N = 1000$). Since a Birkhoff average depends only on the orbit, if the difference in the two answers is small, this is an indication that the average is converging rapidly, and we can classify the orbit as nonchaotic; if the difference is large, the orbit is classified as chaotic. Applying this method to a large number of randomly selected points allows us to identify regions of chaos for billiard maps. These regions are colored gray in all phase plane figures.

Additionally, for nonchaotic orbits, the weighted-Birkhoff average allows us to give a highly accurate computation of the orbit frequency. This method was used to determine the color variation for all phase plane figures.

4.4. **Finding periodic orbits.** In searching for periodic orbits, our goal is to find ones that possess transverse homoclinic points. Therefore, we only seek periodic orbits within regions identified as chaotic.

Initial guesses for the multiple shooting Newton method for periodic orbits can be guessed by looking near elliptical islands. Indeed, since periodic orbits typically appear in hyperbolic

and elliptic pairs, and since the elliptic orbits are easy to see from the coloring of the phase space plot, we can often find initial data for hyperbolic periodic orbits by looking "between" the elliptic islands. Using this geometric intuition, it is possible for orbits of small period, such as say any period up to 5, to simply read good enough initial guesses directly off a phase plane plot colored by frequencies location of chaotic orbits, and to have these guesses converge under Newton's method without further effort. Finding periodic orbits of larger periods is a bit more ad hoc and involves several iterations of the same process, including zooming in on the colored phase plane figure to see more detail of the elliptical islands. With a bit of work we were able to find periodic orbits of periods up to 30. (We did not look for orbits beyond period 30.)

4.5. **Parameterization method for one-dimensional spectral submanifolds.** The material in this section is adapted (in a simplified form appropriate for the present discussion) from [CFdlL03a, CFdlL05]. See also [GMJ17].

Suppose that $p_0 \in \mathbb{R}^d$ is a hyperbolic fixed point for a real analytic map F. In this case there exist $d_u, d_s \in \mathbb{N}$ with $d_u + d_s = d$ such that $DF(p_0)$ has d_s stable eigenvalues $\lambda_1^s, \ldots, \lambda_{d_s}^s \in \mathbb{C}$, and d_u unstable eigenvalues $\lambda_1^u, \ldots, \lambda_{d_u}^u \in \mathbb{C}$. Assume that $DF(p_0)$ is diagonalizable, so that there are $\xi_1^s, \ldots, \xi_{d_s}^s \in \mathbb{R}^d$ associated stable, and $\xi_1^u, \ldots, \xi_{d_u}^u \in \mathbb{R}^d$ unstable eigenvectors. Since $DF(p_0)$ is diagonalizable, these are linearly independent and form a basis for $\mathbb{R}^n$. We are interested in the existence of one-dimensional invariant submanifolds, tangent at p_0 to one of these eigenvectors. Such manifolds are referred to as one-dimensional spectral submanifolds.

One-dimensional submanifolds of the stable/unstable manifold are typically not unique (in fact occur in continua). We select a unique spectral submanifold by imposing maximal regularity. The issue is that, while a typical one-dimensional stable/unstable manifold tangent to an eigenvector is only C^∞ at the fixed point p_0), there is a unique analytic one. The parameterization method is the right tool in this context. It works by looking for an analytic conjugacy to the linear dynamics generated by the eigenvalue. The following lemma provides a single equation which determines a fixed point, its eigendata, and a spectral submanifold. A proof is found for example in [GMJ17].

Lemma 4.2 (Parameterization Lemma)**.** *Let $U \subset \mathbb{R}^d$ be an open, connected set, and $F\colon U \to \mathbb{R}^d$ be a real analytic map. Suppose that $\lambda \in \mathbb{R}$, that $P\colon [-1,1] \to \mathbb{R}^d$ is real analytic on $(-1,1)$, that P is continuous on $[-1,1]$, and that the pair (λ, P) satisfies the conjugacy equation*

$$F(P(\sigma)) = P(\lambda\sigma), \qquad \sigma \in [-1,1]. \tag{9}$$

If $P'(0) \neq 0$ and $0 < |\lambda| < 1$, then:

- *$P(0)$ is a fixed point of F,*
- *$(\lambda, P'(0))$ is an eigenpair for $DF(P(0))$,*
- *$P([-1,1])$ is an analytic curve, tangent to the stable eigenvector associated with λ at $\sigma = 0$, and contained in the local stable manifold of $P(0)$.*

The Lemma explains our interest in Equation (9), but leaves open questions of existence and uniqueness. Here we appeal to Theorem 4.1 of [CFdlL05], which deals precisely with the one dimensional analytic situation at hand (in fact, the theorem is a little more general than we need, as it allows for non-hyperbolicity, complex valued maps, and complex eigendata).

More precisely, assume that the following non-resonance condition holds

$$\lambda^n \notin \mathrm{Spec}(DF(P(0))), \qquad n \geq 2. \tag{10}$$

From this assumption, Theorem 4.1 of [CFdlL05] concludes that Equation (9) has exactly one analytic solution for each choice of eigenvector $P'(0)$.

The choice of the scaling of the eigenvector determines the radius of convergence of P, and by using the rescaling argument from the proof of Theorem 4.1, one can arrange that P is analytic on a disk of radius $1+\epsilon$ in $\mathbb{C}$. If F is real analytic then so is P, and this gives real analyticity on $(-1,1)$ and continuity on $[-1,1]$ as desired. We stress that the proof of Theorem 4.1 gives that the solution of Equation (9) is globally unique as soon as the length of the eigenvector $P'(0)$ is fixed.

We also note, as observed in Remark 4.2 of [CFdlL05], that the invertibility of $DF(P(0))$ implies that there is an $N \geq 2$ so that Equation (10) holds for all $n \geq N$. So, despite first appearances, the non-resonance condition imposes only a finite number of constraints on the eigenvalues of $DF(P(0))$, and Theorem 4.1 applies generically.

The discussion above motivates the following.

Lemma 4.3 (Functional equation, stable case)**.** *Fix a scaling factor $s > 0$ and define the operator $\Psi_s \colon \mathbb{R} \times C^\omega([-1,1],\mathbb{R}^d) \to \mathbb{R} \times C^\omega([-1,1],\mathbb{R}^d)$ by*

$$\Psi_s(\lambda, P(\sigma)) = \begin{pmatrix} \|P'(0)\|^2 - s \\ F(P(\sigma)) - P(\lambda\sigma) \end{pmatrix}$$

If (λ, P) is a zero of Ψ_s with $0 < |\lambda| < 1$, then P satisfies the hypotheses of Lemma 4.2. If the non-resonance condition of Equation (10) is satisfied, then there exists an $\epsilon > 0$ so that for all $0 < s \leq \epsilon$, Ψ_s has a globally unique solution.

Consideration of the unstable manifold leads to a similar zero finding problem with a corresponding but not necessarily equal scaling factor u.

The Fréchet derivatives are written as follows, where $\lambda, \mu, h \in \mathbb{C}$ and $P, Q, H \in C^\omega(\mathbb{B}, \mathbb{R}^d)$.

$$\begin{aligned} D_\lambda \Psi_s(\lambda, P(\sigma))[h] &= \begin{pmatrix} 0 \\ -P'(\lambda\sigma)\sigma h \end{pmatrix}, \\ D_P \Psi_s(\lambda, P(\sigma))[H] &= \begin{pmatrix} 2\langle P'(0), H'(0)\rangle \\ DF(P(\sigma))H(\sigma) - H(\lambda\sigma) \end{pmatrix}, \end{aligned} \tag{11}$$

Remark 4.1 (Generalizations)**.** The parameterization method generalizes to the case when there are resonances. However, one cannot then analytically conjugate to linear dynamics. Instead, one studies a conjugacy equation of the form

$$F(P(\sigma)) = P(K(\sigma)),$$

where K is a polynomial map whose degree and monomial terms are determined by the resonances. More than this, the parameterization method extends to higher dimensional spectral submanifolds, non-hyperbolic fixed points, C^k regularity, and even to other kinds of invariant objects, for both maps and vector fields defined on Banach spaces. Much more general treatment, and many additional references, are found in the paper [CFdlL03a], and also to the book [HCF+18]. See also Appendix B of [CFdlL05] for a scholarly discussion of the historical development of these ideas.

4.6. **Stable and unstable manifolds of a planar hyperbolic periodic orbit.** As an application of the ideas presented in the previous section, we discuss a parameterization method for stable/unstable manifolds of periodic orbits of maps. By restricting to saddle type orbits of planar maps, we obtain an especially complete picture. Note, the material in this section is adapted from [GMJ17], where generalizations to higher dimensional maps and manifolds are also found. We refer also to the paper [TMJ22] where extensions to implicitly defined maps given by polynomial relations are considered.

Let $S \subset \mathbb{R}^2$ be an open and connected set, and suppose that $f\colon S \to \mathbb{R}^2$ is a real analytic mapping. Let us also assume that f is orientation preserving (as is the case for the mathematical billiard maps considered in the present work). Let $F\colon S^K \to \mathbb{R}^{2K}$ be the multiple shooting map for f as defined by Equation (8). Let Ψ_s be the operator defined in Lemma 4.3 where $s > 0$ is fixed. Suppose that for a fixed value of λ with $0 < |\lambda| < 1$ and a fixed function $P\colon [-1, 1] \to \mathbb{R}^{2K}$ we have that $\Psi_s(\lambda, P) = \mathbf{0}$. Combining Lemma 4.1 from Section 4.1 with Lemma 4.2 from Section 4.5, we have the following:

- $P(0) \in \mathbb{R}^{2K}$ is a fixed point of the multiple shooting map F. Letting $P(0) = (p_1, \ldots, p_K)$ with $p_1, \ldots, p_K \in \mathbb{R}^2$, each p_j has period K for f. That is, the components of $P(0)$ provide a periodic orbit for f. If $p_k \neq p_j$ when $j \neq k$, then K is the least period.
- $\lambda^K = \alpha$ is a stable multiplier for the periodic orbit. Moreover, writing $P'(0) = (\xi_1, \ldots, \xi_K)$ with $\xi_1, \ldots, \xi_K \in \mathbb{R}^2$, we have that ξ_j is a stable eigenvector for $Df^K(p_j)$.
- Let $P(\sigma) = (P_1(\sigma), \ldots, P_K(\sigma))$ with each P_j taking values in $\mathbb{R}^2$. Then $P_j(\sigma)$ parameterizes a subset of the stable manifold attached to p_j, and is tangent at zero to the ξ_j.

Since we assume that the orbit is hyperbolic, and that the mapping f is planar, real analytic, and area preserving, we have that the orbit has two real multipliers α, β with $0 < |\alpha| < 1$ and $|\beta| > 1$, and moreover, $\alpha\beta = 1$. Then, they are either both positive, or both negative. In the present work we consider only the case when both multipliers are positive. In this case the eigenvalues of $DF(P(0))$ sort into two sets: the K complex K-th roots of α, and the K complex K-th roots of β. So, every eigenvalue of $DF(P(0))$ has either $|\lambda| = \alpha^{1/K}$ or $|\lambda| = \beta^{1/K}$. Since all the stable eigenvalues have the same (non-unit) modulus, no power of one can equal another, and the non-resonance conditions

$$\lambda^n \notin \mathrm{Spec} DF(P(0)),$$

hold for all $n \geq 2$, and similarly for the unstable eigenvalues.

So, the non-resonance conditions are automatically satisfied and the existence of the desired parameterization is assured. Moreover, since α and β are positive, each has a real K-th root, and associated real eigenvectors. Then the operator Ψ_s defined in Lemma 4.3 has unique, real solutions as long as s remains in the range $0 < s < \epsilon$ for some $\epsilon > 0$. In practice it may not necessarily be the case that ϵ is small, and numerical heuristics are discussed below.

4.7. **Numerical computation of the local parameterizations.** In this section we develop a numerical Newton scheme for the parameterization method which exploits the DFT to compute function compositions. A similar approach is used in [KAdlL25, KAdlL22, KAdlL21] to solve functional equations involving implicitly defined Poincaré maps near periodic solutions in celestial mechanics applications. We review the procedure for the sake of completeness.

Let

$$P(\sigma) = \sum_{n=0}^{\infty} p_n \sigma^n = \sum_{n=0}^{\infty} \begin{pmatrix} p_n^1 \\ \vdots \\ p_n^d \end{pmatrix} \sigma^n, \tag{12}$$

denote a formal power series with $p_n \in \mathbb{R}^d$. Now choose $N \in \mathbb{N}$ and write

$$P^N(\sigma) = \sum_{n=0}^{N} p_n \sigma^n, \tag{13}$$

to denote the truncation of P to order N. We then embed the coefficients of P^N in $\mathbb{R}^{d(N+1)}$ by forming the column vector (written as a row to save space)

$$\mathbf{p} = (p_0^1, \ldots, p_N^1, p_0^2, \ldots, p_N^2, \ldots, p_0^d, \ldots, p_N^d) \in \mathbb{R}^{d(N+1)}. \tag{14}$$

That is, for each $0 \leq n \leq N$, the numbers p_n^i with $1 \leq i \leq d$ are the components of $p_n \in \mathbb{R}^d$.

Now, given $\mathbf{p} \in \mathbb{R}^{d(N+1)}$, we write

$$\mathbf{q} = \mathrm{DFT}[F](\mathbf{p}),$$

to signify that the column vector

$$\mathbf{q} = (q_0^1, \ldots, q_N^1, q_0^2, \ldots, q_N^2, \ldots, q_0^d, \ldots, q_N^d) \in \mathbb{R}^{d(N+1)},$$

is the vectorization of the series coefficients of

$$Q^N(\sigma) = \sum_{n=0}^{N} q_n \sigma^n \approx F(P^N(\sigma)), \qquad |\sigma| < 1,$$

where for each $1 \leq i \leq d$ we compute the series coefficients $\{q_0^i, \ldots, q_N^i\}$ by interpolating the components $F_i \circ P^N$, $1 \leq i \leq d$ of the composition via the DFT interpolation scheme discussed in Appendix B.4.

Similarly, for $1 \leq i, j \leq d$ define the coefficient sequences

$$\mathbf{b}^{ij}(\mathbf{p}) = (b_0^{ij}, \ldots, b_N^{ij}),$$

which result from applying the DFT algorithm discussed in Appendix B.4 to the composition maps $DF_{ij} \circ P$. That is, we let

$$\mathbf{b}^{ij}(\mathbf{p}) = \mathrm{DFT}\left[\partial_j F_i\right](P^N),$$

as these coefficients have

$$\sum_{n=0}^{N} b_n^{ij} \sigma^n \approx \partial_j F_i(P^N(\sigma)) \qquad |\sigma| < 1,$$

where F_i, $1 \leq i \leq d$ denote the component maps of F. We then define the d^2 many $(N+1) \times (N+1)$ matrices

$$B^{ij}(\mathbf{p}) = \mathrm{ConvMat}(\mathbf{b}^{ij}(\mathbf{p})), \tag{15}$$

where the convolution matrix is as defined in Equation (22) of Appendix B.5.

To vectorize the composition $P(\lambda\sigma)$ and its derivatives we first let Λ, Σ denote the $(N+1)\times(N+1)$ diagonal matrices

$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & \lambda & 0 & \dots & 0 \\ 0 & 0 & \lambda^2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda^N \end{pmatrix}, \qquad \text{and} \qquad \Sigma = \begin{pmatrix} 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & N \end{pmatrix},$$

and define $\mathbf{L}_\lambda$ and $\mathbf{N}_N$, the $d(N+1)\times d(N+1)$ diagonal matrices

$$\mathbf{L}_\lambda = \begin{pmatrix} \Lambda & \mathbf{0}_{\mathbb{R}^{N+1}} & \dots & \mathbf{0}_{\mathbb{R}^{N+1}} \\ \mathbf{0}_{\mathbb{R}^{N+1}} & \Lambda & \dots & \mathbf{0}_{\mathbb{R}^{N+1}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{\mathbb{R}^{N+1}} & \mathbf{0}_{\mathbb{R}^{N+1}} & \dots & \Lambda \end{pmatrix}, \qquad \text{and} \qquad \mathbf{N}_N = \begin{pmatrix} \Sigma & \mathbf{0}_{\mathbb{R}^{N+1}} & \dots & \mathbf{0}_{\mathbb{R}^{N+1}} \\ \mathbf{0}_{\mathbb{R}^{N+1}} & \Sigma & \dots & \mathbf{0}_{\mathbb{R}^{N+1}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{\mathbb{R}^{N+1}} & \mathbf{0}_{\mathbb{R}^{N+1}} & \dots & \Sigma \end{pmatrix},$$

and note that $\mathbf{L}_\lambda\mathbf{p}$ and $\mathbf{N}_N\mathbf{p}$ are, respectively, the vectorizations of the coefficients of $P^N(\lambda\sigma)$ and $\sigma(P^N)'(\sigma)$.

Truncating the domain and range of Ψ_s at order N and vectorizing the inputs and outputs leads to the map $\mathbf{\Psi}_s^N\colon \mathbb{R}^{d(N+1)+1} \to \mathbb{R}^{d(N+1)+1}$ given by

$$\mathbf{\Psi}_s^N(\lambda, \mathbf{p}) = \begin{pmatrix} (p_1^1)^2 + \ldots + (p_1^d)^2 - s \\ \\ \mathrm{DFT}[F](\mathbf{p}) - \mathbf{L}_\lambda\mathbf{p} \end{pmatrix}. \tag{16}$$

To vectorize the derivative, we recall the formulas for the action of the Fréchet derivatives given in Equation (11). First, for $1 \le n \le N$ let e_n^* denote the $N+1$ dimensional row vector with a one in the n-th entry and zeros elsewhere. Keep in mind that we start with a zeroth element. Therefore the row vector e_1^* is given by

$$e_1^* = (0, 1, 0, \dots, 0).$$

So, if

$$H^N(\theta) = \sum_{n=0}^{N} h_n \sigma^n,$$

with $h_n \in \mathbb{R}^d$, then the first component of the partial derivative $D_P\Psi_s(\lambda, P)[H]$ given in Equation (11) is

$$\begin{aligned} 2\left\langle (P^N)'(0), (H^N)'(0) \right\rangle &= 2\left(p_1^1 h_1^1 + \ldots + p_1^d h_1^d\right) \\ &= \begin{pmatrix} 2p_1^1 e_1^* & | & 2p_1^2 e_1^* & | & \dots & | & 2p_1^d e_1^* \end{pmatrix} \begin{pmatrix} h_0^1 \\ \vdots \\ h_N^1 \\ \vdots \\ h_0^d \\ \vdots \\ h_N^d \end{pmatrix}. \end{aligned}$$

Suppressing the action on H^N, the partial derivative is represented simply by the $1 \times d(N+1)$ row vector

$$n(\mathbf{p}) = \left(\begin{array}{ccccc} 2p_1^1 e_1^* & | & 2p_1^2 e_1^* & | & \dots & | & 2p_1^d e_1^* \end{array} \right).$$

Similarly, the nontrivial component of the partial derivative $D_\lambda \Psi_s(\lambda, P)$ given in Equation (11) is the negative of $(P^N)'(\lambda\sigma)\sigma$, has n-th coefficient $\lambda^{n-1} n p_n$, and this has vector representation as the $d(N+1)$ dimensional column vector

$$\mathbf{c}_\lambda(\mathbf{p}) = \lambda^{-1} \mathbf{N}_N \mathbf{L}_\lambda(\mathbf{p}).$$

For the second component of partial derivative $D_P \Psi_s(\lambda, P)[H]$, following Equation (11), we vectorize $DF(P^N)$ as follows. Let $\mathbf{B}(\mathbf{p})$ denote the $d(N+1) \times d(N+1)$ matrix

$$\mathbf{B}(\mathbf{p}) = \begin{pmatrix} B^{00}(\mathbf{p}) & \dots & B^{0d}(\mathbf{p}) \\ \vdots & \ddots & \vdots \\ B^{d0}(\mathbf{p}) & \dots & B^{dd}(\mathbf{p}) \end{pmatrix},$$

where for $1 \le i, j \le N$ the $B^{ij}(\mathbf{p})$ are the $(N+1) \times (N+1)$ convolution matrices defined in Equation (15). So, the derivative of $\mathbf{\Psi}_s^N$ has matrix representation

$$D\mathbf{\Psi}_s^N(\mathbf{p}) = \begin{pmatrix} 0 & n(\mathbf{p}) \\ -\mathbf{c}_\lambda(\mathbf{p}) & \mathbf{B}(\mathbf{p}) - \mathbf{L}_\lambda(\mathbf{p}) \end{pmatrix}.$$

Now consider the special case where $F = F_K$ is the multiple shooting map defined in Equation (8), with underlying planar map f, so that $d = 2K$. Let $\bar{\mathbf{p}}_0 \in \mathbb{R}^{2K}$ be an approximate fixed point of the multiple shooting map, in which case the planar components of $\bar{\mathbf{p}}_0$ give an approximate period K orbit for f. Suppose that these components are distinct so that K is the least period.

Assume that $(\bar{\lambda}, \bar{\mathbf{p}}_1)$ is a (numerically computed approximate) real stable (or unstable) eigenpair for $DF(\bar{\mathbf{p}}_0)$. Fix s (or u) a nonzero real number, and rescale $\bar{\mathbf{p}}_1$ so that $\|\bar{\mathbf{p}}_1\|^2 = s$ (or u). Let

$$P^0(\sigma) = \bar{\mathbf{p}}_0 + \sigma \bar{\mathbf{p}}_1. \tag{17}$$

If the data $\bar{\lambda}, \bar{\mathbf{p}}_0, \bar{\mathbf{p}}_1$ were exact, then P^0 would be a quadratically good approximate solution of $\Psi_s = 0$. That is, we would have

$$\|\Psi_s(P^0)\| \le C s^2.$$

So, assuming that $\bar{\mathbf{p}}_0$ and $\bar{\mathbf{p}}_1$ are good enough approximations, we expect to be able to find an $s, u > 0$ so that this initial error is small (limited only by the error in the initial data which is, presumably, on the order of machine epsilon).

Let $\mathfrak{p}^0$ be the vectorization of $P^0(\sigma)$, and define the Newton sequence

$$\mathfrak{p}^{m+1} = \mathfrak{p}^m + \mathbf{\Delta}^m,$$

where, for $m \ge 0$, $\mathbf{\Delta}^m$ solves the linear equation

$$D\mathbf{\Psi}_s^N(\mathfrak{p}^m)\mathbf{\Delta}_m = -\mathbf{\Psi}_s^N(\mathfrak{p}^m).$$

In practice we experiment with the scalings s, as we would like to choose the largest such values so that the Newton sequence converges. More sophisticated continuation schemes and optimization algorithms for choosing computational parameters for the parameterization method are discussed in [BLMJ16, HMJ21].

Remark 4.2 (Automatic reducibility)**.** Note that the Newton method just described is a "large matrix method". More precisely, if K is the period of the orbit, and N is the polynomial order of approximation, then the linear system determining the Newton step has size $(2K(N+1)+1) \times (2K(N+1)+1)$. While this presents no problem in the present work, the complexity can be greatly reduced by exploiting the symplectic structure of the billiard system to define a diagonal approximate inverse for the Newton matrix. This is referred to as "approximate reducibility" in the literature, and the interested reader will find much more on this topic, including many references to its use in the literature, in the book [HCF+18]. Using approximate reducibility would be helpful for studying orbits with much higher period than those studied here.

Remark 4.3 (The method of jet transport)**.** We note that an alternative to the DFT based interpolation scheme described above would be to compute necessary compositions using the method of jet transport described in [GJJCZ25]. Indeed, the reference just cited uses jet transport to solve equations similar to the conjugacy equations considered in the present work, applying these ideas to Poincaré maps for ODEs in celestial mechanics. These implicit Poincaré maps are, if anything, more complicated than the billiard maps studied in the present work, so we assume that the method of jet transport could be applied to billiards as well. The only disadvantage of jet transport is that it requires special software like the TAYLOR package developed by the authors of the reference just cited. Our implementation relies only on standard numerical linear algebra routines like linear system solvers, eigenvalue/eigenvector solvers, and (if desired) the FFT. These algorithms are available in any standard numerical linear algebra package. Our codes for example are implemented in MATLAB.

A useful a-posteriori error indicator is obtained by measuring the defect in the conjugacy relation. So, define $E_{\text{conj}} \colon \mathbb{R} \times C^\omega(\mathbb{D}, \mathbb{R}^{2K}) \to \mathbb{R}$ by

$$E_{\text{conj}}(\lambda, P) = \sup_{\sigma \in \mathbb{D}} \|F(P(\sigma)) - P(\lambda\sigma)\|_{\mathbb{R}^{2K}}$$

Note that $E_{\text{conj}}(\bar{\lambda}, \bar{P}) = 0$ if and only if $(\bar{\lambda}, \bar{P})$ is the desired solution. A numerical proxy for the a-posteriori indicator is obtained by sampling the right hand side at $M \in \mathbb{N}$ points on the boundary of $\mathbb{D}$, and computing the maximum of this finite set. More precisely, by the maximum modulus principle we have that

$$E_{\text{conj}}(\lambda, P) \approx E^M_{\text{conj}}(\lambda, P) = \max_{0 \le m \le M} \left\| F\left(P\left(e^{\frac{2\pi i m}{M+1}}\right)\right) - P\left(\lambda e^{\frac{2\pi i m}{M+1}}\right)\right\|_{\mathbb{R}^{2K}},$$

bounds $E_{\text{conj}}(\lambda, P)$ from below, is computable in finitely many operations, and converges to the true a-posteriori indicator as $M \to \infty$. Note also that, in the case that $P = P^N$ is polynomial, it is efficiently evaluated at the desired roots of using the FFT.

4.8. **Growing the local manifolds.** To compute larger portions of the stable and unstable manifolds, we employ a fundamental domain approach that leverages the periodic structure of the system. Let $\mathcal{W}^s$ and $\mathcal{W}^u$ denote the global stable and unstable manifolds, respectively.

We work within a fundamental domain $\mathcal{D} \subset \mathbb{R}^2$, a bounded region such that every point on the global manifold can be obtained by applying forward or backward iterates of the map F to points in $\mathcal{D}$.

Given the local parameterizations $P^s(\theta)$ and $P^u(\theta)$ of the stable and unstable manifolds, the global manifolds are constructed by first selecting parameterization points θ_j such that

$P^{s,u}(\theta_j) \in \mathcal{D}$. Then for each selected point $x_j = P^{s,u}(\theta_j)$, where $j = 1, \ldots, J$, compute:

$$\mathcal{W}^u_{\text{global}} \approx \bigcup_{j=1}^{J} \bigcup_{n=0}^{M} F^n(x^u_j) \tag{18}$$

$$\mathcal{W}^s_{\text{global}} \approx \bigcup_{j=1}^{J} \bigcup_{n=0}^{M} F^{-n}(x^s_j)$$

where M is the maximum number of iterates. This approach ensures that the computed global manifolds cover the relevant phase space region while avoiding redundant computation outside the fundamental domain.

5. Results

In this section, we give our results. As mentioned in the introduction, the importance of our findings is twofold: the novelty of the use of the method, and also the implications for billiard maps. Correspondingly, we divide this section into two parts.

5.1. **Results for the parameterization method.** We compute the stable and unstable manifolds for billiard maps using the methods described in Section 4.7. The two crucial sets of parameters are the truncation order of the polynomial approximation N, introduced in Equation (13) and the eigenvector scaling. The scaling coefficients are labeled s in Lemma 4.3, but in this section and in our Tables of numerical parameters, we refer to these parameters by scale_s and scale_u to distinguish these cases. The resulting local manifolds are shown in the left panels of Figures 9–11 for the period-10 orbit for Table D, the period-3 orbit for Table C, and the period-2 orbit for Table B (cf. Tbl. 1), respectively. The corresponding global manifolds are obtained by iterating the local parameterizations under F. The right panels of the same figures illustrate these global structures, together with the truncation order N of the polynomial approximation $P^N(z)$ and the number of iterates used.

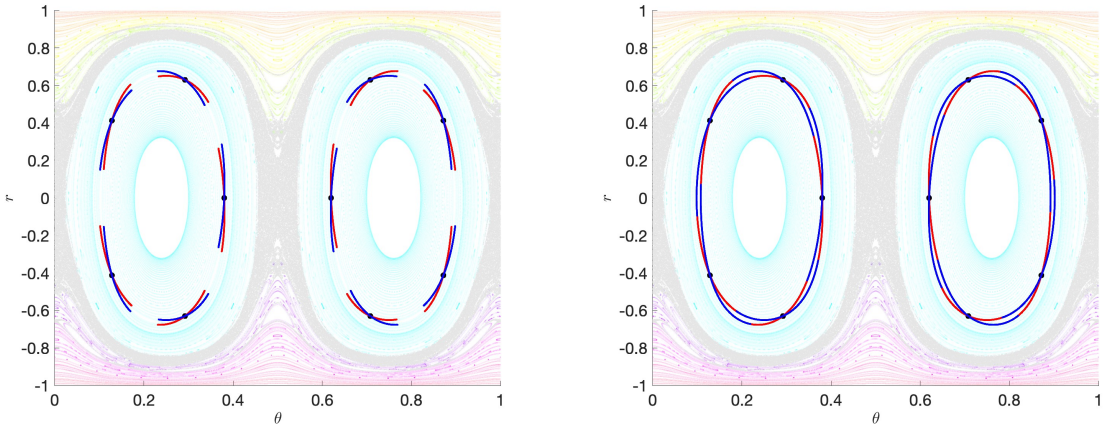


FIGURE 9. Local (left) and global (right) stable (blue) and unstable (red) manifolds for a period-10 orbit for Table D (cf. Tbl. 1). The global manifolds were computed with 20 iterates of the fundamental domain and $N = 60$.

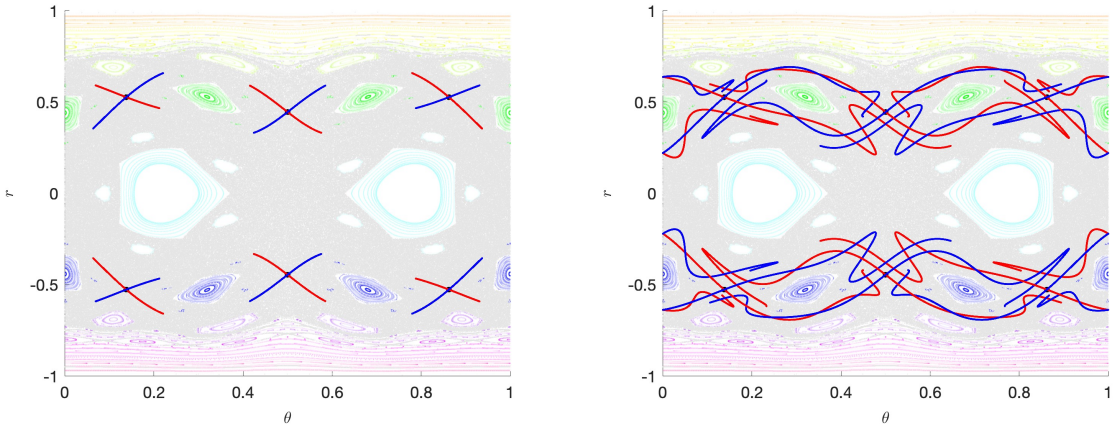


FIGURE 10. Local (left) and global (right) stable (blue) and unstable (red) manifolds for a period-3 orbit for Table C (cf. Tbl. 1). The global manifolds were computed with 6 iterates of the fundamental domain and $N = 60$.

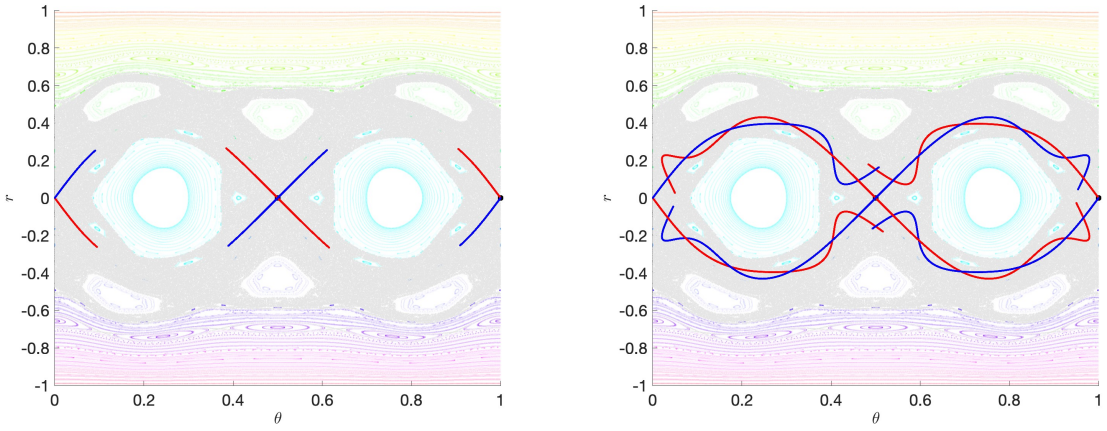


FIGURE 11. Local (left) and global (right) stable (blue) and unstable (red) manifolds for a period-2 orbit for Table B (cf. Tbl. 1). The global manifolds were computed with 4 iterates of the fundamental domain and $N = 60$.

The novelty of the present computation lies in performing the Newton iteration directly in the spectral domain, allowing both stable and unstable parameterizations to be obtained with high-order accuracy and without explicit recursion on the coefficients. In contrast to traditional recursive or pointwise approaches, this formulation combines spectral evaluation and FFT-based transforms to achieve rapid convergence, ensuring both robustness and precision in the manifold computation.

To verify the spectral accuracy of the computed parameterizations, we analyze the decay of the coefficients a_n of P^N obtained from the Newton iteration. For both the stable and

unstable manifolds, the magnitude of each coefficient is evaluated as $\|a_n\|_\infty$, and its logarithmic scale $\log_{10}(\|a_n\|_\infty)$ is plotted against the mode index n. The resulting coefficient profiles, shown in the center and right panels of Figure 12 for a period-5 orbit of Table B (cf. Tbl. 1), exhibit a rapid exponential decay of the coefficients, reaching values of order 10^{-16} by the truncation index N. This confirms that the spectral representation $P^N(z) = \sum_{n=0}^{N} a_n z^n$ provides a highly accurate analytic approximation of the local manifolds within the domain of convergence of the parameterization.

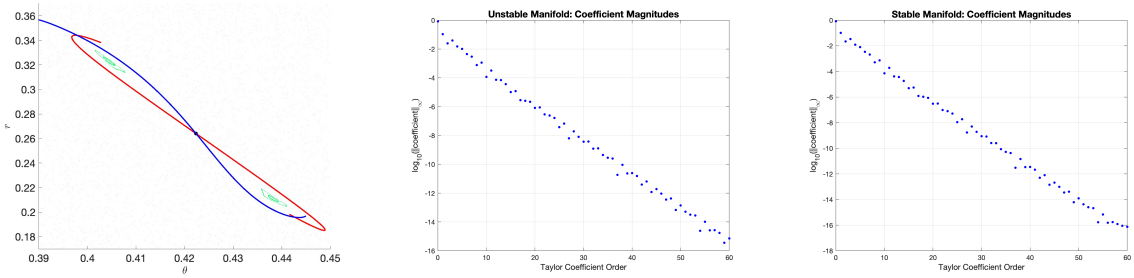


FIGURE 12. (Left) Computed global stable and unstable manifolds for one point of the period-5 orbit of Table B (cf. Tbl. 1). (Center, right) Logarithmic decay of the infinity norm of Taylor coefficients $\|\mathbf{a}_n\|_\infty$ for the local unstable and stable manifolds, respectively. Computations performed using $N = 60$ and 5 iterates of the fundamental domain.

Another key advantage of our approach is that it does not stop at the linear tangent eigenvector chart, but by increasing the truncation order N of the expansion of $P^N(z)$, one can capture the nonlinear deformation of the invariant manifold at larger amplitude. In particular, while the leading term a_1 corresponds to the linear eigenvector scaling, the higher coefficients a_n with $n \geq 2$ encode the curvature, folding and higher-order geometry of the manifold. By systematically computing and monitoring these coefficients, our implementation allows the eigenvector to be scaled to increasingly large magnitudes and hence captures local nonlinear behavior of the manifold near the periodic orbit, see Figures 13-15 for a visual illustration of this behavior. In both figures, increasing N captures more nonlinear geometry of the manifolds, with scale parameters scale_u and scale_s (see Appendix A.2 for definitions) adjusted accordingly to control the extent of the parameterization from the periodic orbit.

To assess the robustness of our numerical methods, we systematically increase the perturbation amplitude applied to the coefficients in Table C (cf. Tbl. 1). Specifically, we multiply all coefficients by a factor of 3.5, representing an extreme perturbation far beyond the nominal parameter regime. In this strongly perturbed case, the phase space reveals extensive chaotic behavior: iterates densely populate the entire accessible domain in the (θ, r) plane, forming a nearly uniform cloud of points that obscures much of the underlying structure. The chaotic sea extends throughout the region $0 \leq \theta \leq 1$ and $-0.5 \leq r \leq 0.5$, with particularly dense regions near $r \approx \pm 0.5$ where iterates exhibit wild oscillations. Despite this extreme perturbation—a 250% increase over the nominal values—we successfully compute the stable and unstable manifolds of the hyperbolic periodic-2 orbits located near $(\theta, r) \approx (0.25, 0)$ and $(0.75, 0)$. The successful resolution of these invariant manifolds under such severe perturbation, where the system dynamics have deteriorated into an almost completely chaotic state,

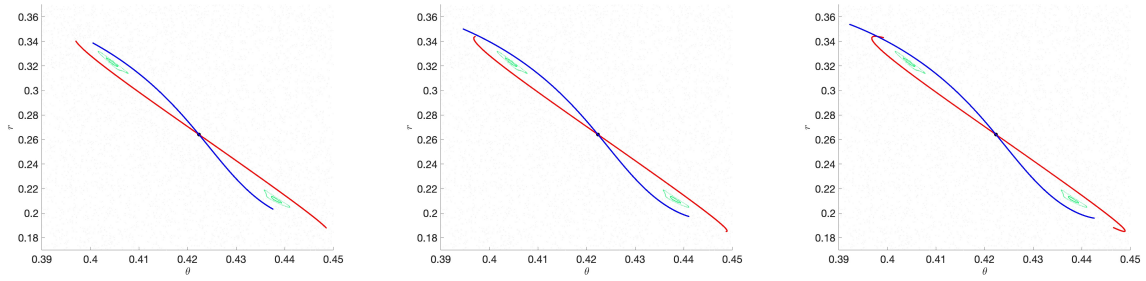


FIGURE 13. Computed local stable and unstable manifolds for the same period-5 orbit from Figure 12 with varying polynomial order N. (Left) $N = 60$, $\text{scale}_u = 0.21$, $\text{scale}_s = 0.2$. (Center) $N = 100$, $\text{scale}_u = 0.26$, $\text{scale}_s = 0.26$. (Right) $N = 180$, $\text{scale}_u = 0.32$, $\text{scale}_s = 0.293$; transverse intersections occur even for the local manifolds.

demonstrates the remarkable effectiveness of our computational approach even as it operates near the boundary of numerical feasibility.

As a further robustness check, we tested a boundary perturbation with many small nonzero Fourier modes rather than only three: we set $a_{k,1} = a_{k,2} = \nu/k^2$ for $k = 1, \ldots, 20$, with $\nu = 0.03$ matching the leading-mode amplitude of Table A, keeping $a_{1,1} = 1.1$ and $b_{1,2} = 1$ as in Tables A–C. We computed the local stable and unstable manifolds of a period-5 orbit for this perturbation. Due to the slow decay, a larger proportion of the phase space is chaotic, and the resulting manifolds are more oscillatory than the ones we found for Tables A–E, see Figure 14. The periodic orbit was refined to a closure error of 3.318×10^{-16}, with multipliers $\lambda_u = 1.558838$ and $\lambda_s = 0.641503$, consistent with a hyperbolic saddle orbit. The Newton iteration for the stable manifold converged to the 10^{-14} tolerance in 6 iterations, while the unstable manifold converged to a residual of 2.24×10^{-12} after 10 iterations. The corresponding Taylor coefficients decay approximately exponentially over the full range $0 \leq N \leq 60$, reaching magnitudes below 10^{-12}, comparable to the decay observed for Tables A–E (cf. Figure 12). These results indicate that the efficiency and accuracy of our method are not significantly affected by the presence of many small nonzero Fourier coefficients, as opposed to a single dominant mode. We additionally benchmarked the per-call runtime of the map and its Jacobian for a fixed polynomial across a family of decay laws ($1/k^p$ for $p = 3, \ldots, 7$ and $1/2^k$) and mode counts $N = 2, 3, 10, 20, 30, 40, 60, 80, 100$, evaluated at a fixed, Newton-refined periodic orbit point; runtime scales approximately linearly with the number of retained modes once the boundary geometry stabilizes, and is not sensitive to the specific choice of decay law.

5.2. **Results for billiards.** Here we concentrate on observations in terms of what our manifolds indicate about the billiard tables. In Appendix A, we have listed all of the periodic orbits found for the five tables. For example, for Billiard Tables A and B, we found a total of eight periodic orbits and manifolds, with periods 2, 3, 5, 10, and 30, with a similar scope in all other cases. In Figures 2–4, we give an overview of all of our computations of invariant manifolds of Tables A–E listed in (2) and Tbl. 1.

In most of the examples we computed, we find that either the local manifold already intersect transversely, or that this clearly happens after a small number of iterates. A notable

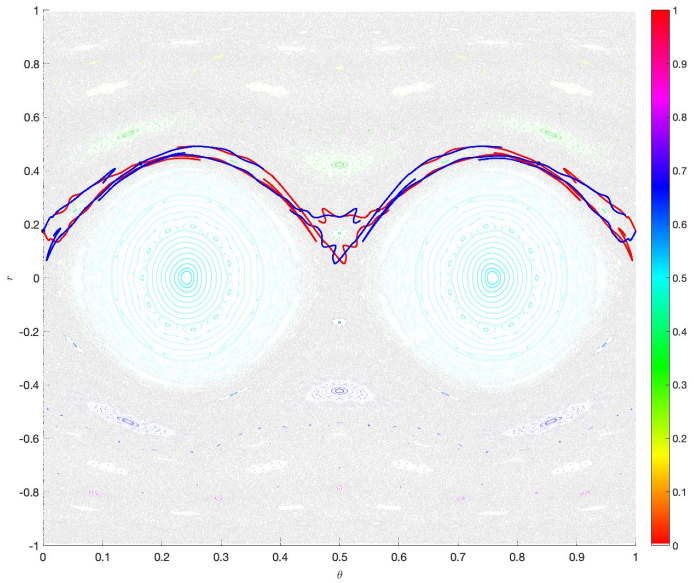


FIGURE 14. Stable (blue) and unstable (red) manifolds of the period-5 orbit for the $1/k^2$-decaying, $N = 20$-mode boundary perturbation, overlaid on the phase plane (θ, r).

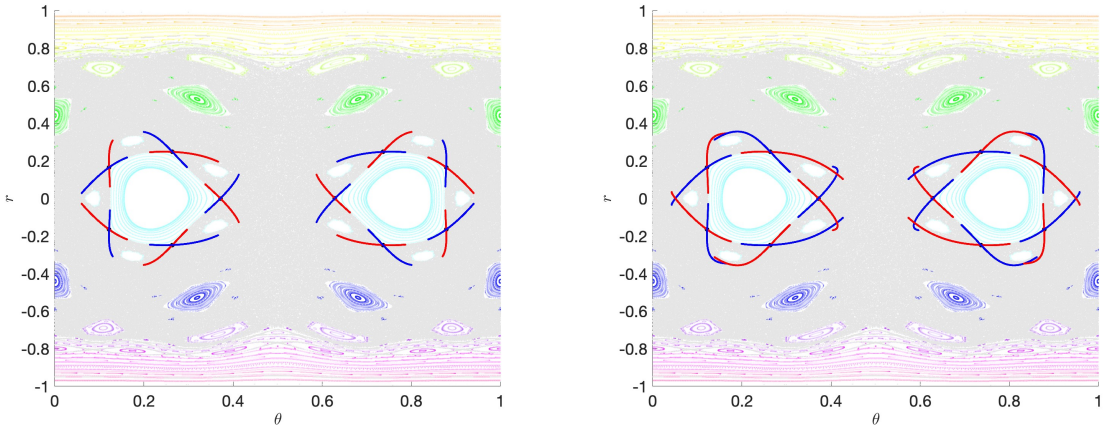


FIGURE 15. Computed local stable and unstable manifolds for the period-10 orbit for Table C (cf. Tbl. 1) with varying polynomial order N. (Left) $N = 60$, $\text{scale}_u = 0.34$, $\text{scale}_s = 0.34$. (Right) $N = 200$, $\text{scale}_u = 0.51$, $\text{scale}_s = 0.51$; transverse intersections occur even for the local manifolds.

exception is the period-30 case, where the splitting angle appears to be very small. This is not unexpected, as the splitting angle is known to be exponentially small in the period. For the period 30 orbit on Table A, the intersection only becomes visible once the number of iterates

is increased to 100, see Figure 16. For Table B it remains unresolved whether the stable and unstable manifolds for period-30 intersect transversely even after 100 iterates. More generally, due to exponential splitting, we expect that it would not be possible to distinguish transverse intersections for many high period orbits without using extended-precision calculations, such as the ones performed in [MRRTS16].

Due to the homoclinic connections, the manifolds are trapped inside the connected chaotic region containing the original periodic orbit, wrapping around to avoid the islands. For example, see the period-2 manifolds wrapping around the two main islands in Figure 11, the period-5 manifolds wrapping around a small pair of islands in Figure 12, and the period-10 manifolds wrapping around two star-shape pieces consisting of twelve islands in Figure 15. The larger eccentricity cases have the same behavior, but all the islands are squeezed into a smaller space, meaning that it is harder to observe.

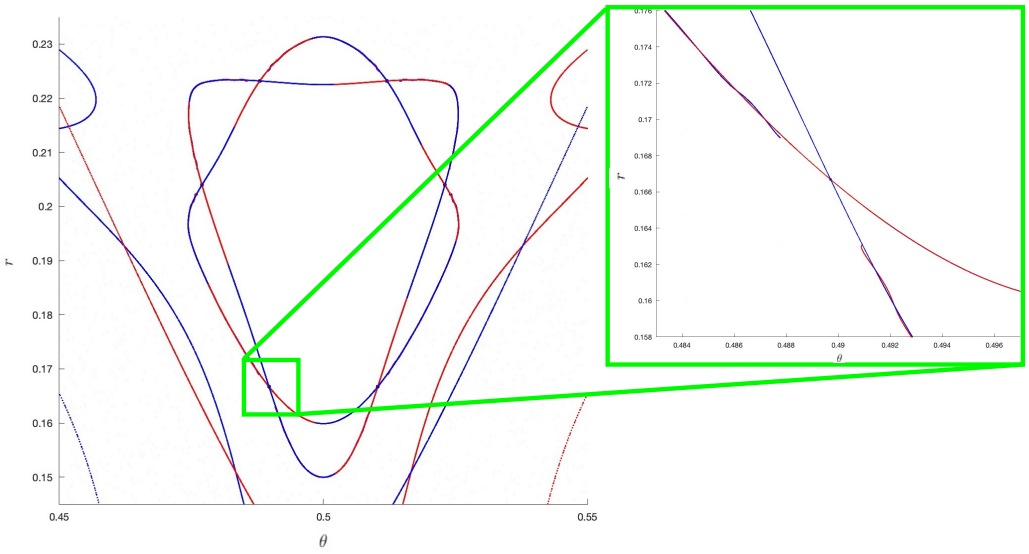


FIGURE 16. Computed global stable and unstable manifolds for a six point set of the period-30 orbit for Table A (cf. Tbl. 1), with 100 iterations of the fundamental domain. (Right) Zoomed plot at one of the period-30 points, showing the transverse intersection.

The resulting manifolds and homoclinic tangle beautifully align with the chaotic regions computed using the weighted Birkhoff average. Each set of manifolds and their intersections fills up the entire connected chaotic sea that contains it, giving good agreement between these two methods. This is already somewhat clear in Figures 2-4, although we had kept the number of iterations rather small to avoid overcrowding. It is much more obvious in Figure 17, where we show only one set of manifolds and allow the number of iterations of the local manifolds to be ten.

We are certainly not the first to observe that these maps bear a striking resemblance to the behavior of the Chirikov standard map. In the perturbed case (positive parameter for the Chirikov standard map), they both have elliptic islands, along with small chaotic regions, which are filled with homoclinic tangles of hyperbolic periodic orbits. For a strictly convex billiard table, if the boundary map B is parameterized by arclength, then the billiard

map is symplectic [Mei92]. Though we do not assume arclength parameterization, since the frequency is independent of the parameterization, we can conclude from this that the frequency of the rotational tori increases as r increases, as we can observe from the figures of the phase space.

We note that there are a few key differences. The Chirikov standard map has large elliptic period-one islands, whereas here the lowest period possible is two. For the Chirikov standard map, there is a famous parameter value for which all rotational orbits break up [SM20]. In the billiards case, no perturbation breaks up all rotational orbits. In particular, for every table with a C^6 smooth strictly convex boundary, there exist "whispering gallery" orbits [Laz73, Dou82]. Finally, the Chirikov standard map is entire, whereas billiard maps can only be continued to a finite thickness strip of the complex plane. Of particular interest is how well

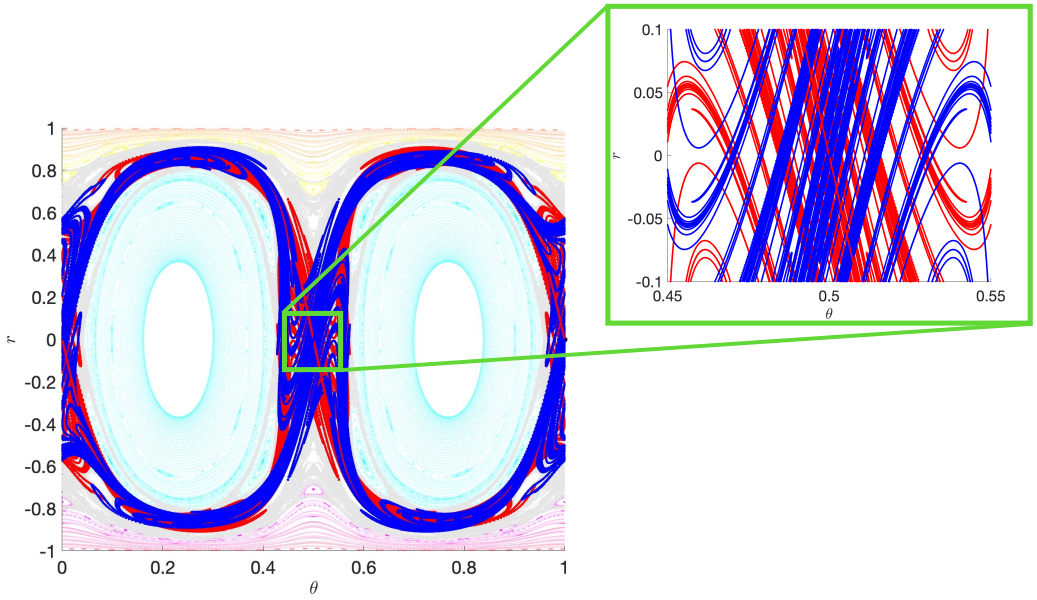


FIGURE 17. (Left) Computed global stable and unstable manifolds for a period-2 orbit for Table E (cf. Tbl. 1), $N = 40$, $\text{scale}_u = 0.52$, $\text{scale}_s = 0.48$ and 10 iterations of the fundamental domain. (Right) Zoomed in plot at one of the period-2 points $(\theta, r) = (0.5, 0)$ to clearly discern chaotic behavior in that particular region.

we can see homoclinic tangle within the chaotic region, such as the Cantor set type behavior of the manifolds, and how well it illustrates in this context the results of the λ-lemma.

The λ–lemma describes a fundamental geometric mechanism by which images of disks (or arcs) transverse to a stable (resp. unstable) manifold are stretched and asymptotically aligned with the unstable (resp. stable) manifold under forward (resp. backward) iteration of the dynamics; informally, a small transverse disk is "carried along" and accumulates onto the invariant manifold while its tangent directions converge to the tangent bundle of that manifold. This statement is a classical ingredient in hyperbolic dynamics and its ramifications for homoclinic and heteroclinic behavior are well documented (see, [Rob98, CFJ20]).

Numerically, the mechanism predicted by the λ–lemma is captured here in two complementary ways. First, the high-order parameterizations $P^N(z)$ provide accurate local charts (disks) transverse to the periodic orbit together with reliable approximations of their tangent bundles: the leading coefficient a_1 recovers the tangent eigenvector while the higher coefficients $a_{n\geq 2}$ encode the local nonlinear deformation. Second, by iterating the computed local chart under the map F (or its inverse) and monitoring both pointwise images and the evolution of tangent information, we observe the characteristic stretching, alignment and accumulation described by the λ–lemma. Practically, the method's spectral accuracy (exponential coefficient decay to machine precision) and the Newton method applied ensure that these iterated images remain within the radius of analyticity of the parameterization and that the observed alignment is not a numerical artifact. Thus, by combining high-order spectral charts with controlled iteration in the fundamental domain, our implementation reproduces the geometric content of the λ–lemma in a numerically verifiable way. A visualization of this phenomenon can be seen in Figure 17 for a period-2 orbit of Table E (cf. Tbl. 1).

6. Conclusions and future plans

In this paper, we developed a C^1 method for iterating billiard maps for tables with smooth convex boundaries. We then used the weighted-Birkhoff average to distinguish between chaos and quasiperiodic orbits, at the same time as computing the frequency of the quasiperiodic orbits. We used the method of multiple-shooting to find periodic orbits for the billiard map and selected the hyperbolic periodic orbits. We then used the parameterization method to compute the local stable and unstable manifolds for the periodic orbits of periods 2, 3, 4, 5, 10, and 30. However, our methods could also be extended to compute for other lower and higher periods. This use of the parameterization method is novel, in that the billiard map is fully implicit. Therefore, to find the coefficients of the Taylor series approximation of the manifolds, we analytically continued our billiard map into the complex plane, allowing us to compute series coefficients using the Discrete Fourier Transform. Finally, we are able to approximate the global manifolds up to a chosen number of iterations and find transverse intersections.

In addition to the computational novelty of this work, the application to billiards is itself of great interest. Previous results on homoclinic tangles for billiard maps have concentrated on the singular limit as the perturbation approaches the ellipse, showing that there are exponentially small angles of intersection of the manifolds. In contrast, we are interested in homoclinic tangles for billiard tables which are non-limiting perturbations from the ellipse.

One of the advantages of the parameterization method for computing manifolds is that in addition to giving high accuracy approximations, the method can be used for computer assisted proofs. Not just local manifolds, but also a finite number of iterations of these local manifolds can be rigorously validated. In future work, we plan to validate our manifolds. Since the existence of transverse homoclinic points implies chaos, such a validation will lead to a rigorous validation of chaos for perturbed elliptic billiard maps.

The numerical algorithms described in this work have been implemented in MATLAB and are available as open-source code at https://github.com/efleurantin2103/Billiards, enabling reproduction of all computational results presented here.

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Appendix A. Computational Parameters and Technical Details

This appendix provides the parameters used to compute the global stable and unstable manifolds. It also gives a technical modification for our numerical method for finding an initial guess for f.

A.1. **Modification to find our initial guess.** We now describe our second technical modification to our method for computing the initial guess in order to perform Newton's method to find the billiard iterate $f(\theta, r)$. To make sure that the initial guess is always well defined, we consider $\mathbf{v}_{-1}$, the previous direction of $\mathbf{v}$, and use the $\mathbf{v}_e$ that is the reflection of $\mathbf{v}_{-1}$ off the elliptical table.

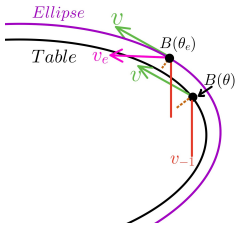


Figure 18. The reflection vector v for our table might not point into the interior of the associated elliptical table. To make sure it is always well defined, we use v_e in finding the guess $(\hat{\theta}_e, \hat{r}_e)$.

A.2. We now present the parameters and values of the periodic orbits seen in Figures 2–4 in a series of data Tables, one for each period. More specifically, each data Table corresponds to orbits of a specific period and references the billiard tables in the main text where these orbits are analyzed. The parameter r represents the radial coordinate, θ denotes the parameter describing the point on the boundary of the billiard table, and scale_u and scale_s are the scaling factors applied to the unstable and stable eigenvectors, respectively, in the parameterization method. (These were labeled u and s respectively in the theoretical discussion in Section 4.5.) The variable N indicates the order of the polynomial approximation, while iterations refer to the number of iterates M used in the computational scheme. These parameters were carefully chosen to ensure convergence and accuracy of the computed invariant manifolds associated with each periodic orbit.

Appendix B. Numerical manipulation of power series

Consider a power series

$$P(\sigma) = \sum_{n=0}^{\infty} a_n \sigma^n,$$

with $a_n \in \mathbb{R}$ (or $\mathbb{C}$), and let $g\colon \mathbb{C} \to \mathbb{C}$ be an analytic function. We seek the power series coefficients of

$$(g \circ P)(\sigma) = \sum_{n=0}^{\infty} b_n \sigma^n.$$

These are given explicitly by the Faà di Bruno formula (multivariate generalization of the chain rule). This however leads to exponential complexity, and we consider alternatives.

Period 2	r	θ	scale_u	scale_s	Iterations	N
Table A	0.0000	0.5000	0.48	0.46	6	60
Table B	0.0	0.5	0.45	0.43	5	60
Table C	0.0	0.5	0.32	0.32	6	60
Table D	0.0	0.5	0.12	0.1	2	10
Table E	0.0	0.5	0.12	0.12	2	10
Period 3	r	θ	scale_u	scale_s	Iterations	N
Table A	0.5333	0.8534	0.25	0.25	10	60
Table A	-0.5333	0.8534	0.25	0.25	10	60
Table B	0.5328	0.3348	0.27	0.27	7	60
Table B	-0.5328	0.3348	0.27	0.27	7	60
Table C	0.4454	0.5	0.28	0.28	6	60
Table C	-0.4454	0.5	0.28	0.28	6	60
Period 4	r	θ	scale_u	scale_s	Iterations	N
Table C	0.7078	0	0.2	0.2	8	60
Table C	-0.7334	0.2152	0.2	0.2	8	60
Table D	0.5	0	0.114	0.114	8	35
Table D	-0.8923	0.2216	0.114	0.114	8	35
Table E	0.89	0.22	0.14	0.115	3	25
Table E	-0.89	0.22	0.14	0.115	3	25

TABLE 2. Numerical parameters for the orbits of periods 2, 3, and 4. The value N is the order of the polynomial approximation given in (12), Iterations is the number M of iterates in (18), and scale_s and scale_u are the scalings of the eigenvectors in the parameterization method defined in Lemma 4.3.

Period 5	r	θ	scale_u	scale_s	Iterations	N
Table A	0.2585	0.4308	0.29	0.29	11	60
Table A	-0.2585	0.4308	0.29	0.29	11	60
Table B	0.2641	0.4223	0.21	0.2	10	60
Table B	-0.2641	0.4223	0.21	0.2	10	60
Table C	0.7695	0.5	0.17	0.17	25	60
Table C	-0.7695	0.5	0.17	0.17	25	60
Table D	0.75	0.44	0.125	0.123	10	30
Table D	-0.75	0.44	0.125	0.123	10	30
Table E	0.7	0.46	0.125	0.123	7	25
Table E	-0.7	0.46	0.125	0.123	7	25
Period 10	r	θ	scale_u	scale_s	Iterations	N
Table A	-0.3283	0.701	0.39	0.39	10	60
Table B	-0.1746	0.3767	0.46	0.48	4	60
Table C	0.1653	0.122	0.34	0.34	8	60
Table D	0	0.3802	0.7	0.7	20	60
Table E	-0.4106	0.1275	0.65	0.65	20	60
Period 30	r	θ	scale_u	scale_s	Iterations	N
Table A	0.1667	0.5103	0.15	0.15	20	60
Table A	-0.1667	0.4897	0.15	0.15	20	60
Table B	0.4877	0.5	0.15	0.15	30	60
Table B	-0.4877	0.5	0.15	0.15	30	60

TABLE 3. Numerical parameters for the orbits of periods 5, 10, and 30.

B.1. **Polynomial case.** A common situation is that g is polynomial, in which case explicit recursive formulas for the coefficients of integer powers of P are worked out using the Cauchy product. For example we have that

$$P(\sigma)^2 = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_{n-k} a_k \right) \sigma^n, \tag{19}$$

and that

$$P(\sigma)^3 = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} \sum_{j=0}^{k} a_{n-k} a_{k-j} a_j \right) \sigma^n.$$

In these cases

$$b_n = \sum_{k=0}^{n} a_{n-k} a_k, \qquad \text{and} \qquad b_n = \sum_{k=0}^{n} \sum_{j=0}^{k} a_{n-k} a_{k-j} a_j,$$

respectively. Formulas for higher order powers generalize in an obvious way. Therefore Cauchy products are sufficient for working out the power series coefficients for $g \circ P$ in the polynomial case.

B.2. **D-finite functions: differential algebra.** If g is non-polynomial, but elementary (sometimes referred to as D-finite), then the coefficients of $g \circ P$ are often found by combining the fact that the chain rule turns composition into multiplication with the Cauchy product. For example, suppose $g(z) = e^z$. We write

$$Q(\sigma) = \sum_{n=0}^{\infty} b_n \sigma^n = g(P(\sigma)) = e^{P(\sigma)},$$

to denote the unknown power series of the composition and have that

$$Q'(\sigma) = \sum_{n=0}^{\infty} (n+1) b_{n+1} \sigma^n = g'(P(\sigma)) P'(\sigma) = e^{P(\sigma)} P'(\sigma),$$

which is

$$Q'(\sigma) = Q(\sigma) P'(\sigma).$$

Expressing this last expression in terms of power series and calling on the Cauchy product, we have that

$$\begin{aligned} \sum_{n=0}^{\infty} (n+1) b_{n+1} \sigma^n &= \left(\sum_{n=0}^{\infty} b_n \sigma^n \right) \left(\sum_{n=0}^{\infty} (n+1) a_{n+1} \sigma^n \right) \\ &= \sum_{n=0}^{\infty} \left((k+1) b_{n-k} a_{k+1} \right) \sigma^n. \end{aligned}$$

Matching like powers of σ^n, yields the recursive expression for the power series coefficients of $g \circ P$ given by

$$b_{n+1} = \frac{1}{n+1} \sum_{k=0}^{n} (k+1) b_{n-k} a_{k+1}, \qquad n \geq 0,$$

with

$$b_0 = e^{a_0}.$$

Using this differential-algebraic framework, we find that the power series coefficients of $g \circ P$ are no more difficult to compute in this transcendental case than they are in the quadratic case.

B.3. **Remarks on the literature.** This idea illustrated in the previous section is sometimes referred to as automatic differentiation for power series, polynomial recast, or polynomial embedding, and it is sufficient for many problems with reasonably complicated nonlinearities. An important general reference is Chapter 4.6 of the second volume of Knuth's The Art of Computer Programming [Knu98]. See also [JZ05] for explicit recursion relations for many elementary functions. Indeed, the reference just cited combines automatic differentiation for power series with a state-of-the-art parser as part of a powerful C++ software library called TAYLOR, and suggests that the boundary between simple and complex nonlinearities is fluid, when approached with the right tools. We refer also to the papers [FGJ25, GJJCZ25] where more recent enhancements and applications of the TAYLOR package are discussed. Further extension of these ideas to multivariable power series is discussed at length in Chapter 2.3 of [HCF+18].

This said, there may be (depending on one's tolerance for complexity) problems where explicit formulas for g are either too much to hope for, or are more complicated than one wishes consider (especially if one is unwilling to adopt special purpose software). For example, Poincaré sections for ODEs and billiard maps studied in the present work approach this boundary. For such maps, we have ready access only to numerical procedures that compute the values g (and its derivatives) at points. In this case, interpolation is a useful tool for determining the power series coefficients of $f \circ P$, and when the functions are analytic, the interpolation can be carried out via the discrete Fourier transform (DFT). An excellent reference for this material is the book of [BH95].

B.4. **Composition of power series via the DFT.** Suppose that $P \colon \mathbb{R} \to \mathbb{R}^d$, and $g \colon \mathbb{R}^d \to \mathbb{R}$ are real analytic, $g(P(\sigma)) = Q(\sigma)$, whose power series we approximate with the polynomial interpolant

$$Q^N(\sigma) = \sum_{n=0}^{N} \bar{b}_n \sigma^n.$$

The coefficients $\bar{b}_n \approx b_n$, $0 \leq n \leq N$ of the interpolant are determined by the values of Q at any $(N+1)$ distinct points, and we choose values equidistributed on the unit circle in $\mathbb{C}$ with the first equal to one.

More precisely, define the $N+1$ grid points

$$\sigma_k = e^{2\pi i k/(N+1)}, \quad 0 \leq k \leq N.$$

and evaluate the composition on this mesh to obtain

$$Q^N(\sigma_k) = \sum_{n=0}^{N} \bar{b}_n \sigma_k^n \approx g(P(\sigma_k))$$

(20)

for $0 \leq k \leq N$. Letting $\omega = e^{\frac{2\pi i}{N+1}}$, the b_n are approximated by $\bar{b}_n$ solving the linear system

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^N \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^N & \omega^{2N} & \cdots & \omega^{N^2} \end{pmatrix} \begin{pmatrix} \bar{b}_0 \\ \bar{b}_1 \\ \bar{b}_2 \\ \vdots \\ \bar{b}_N \end{pmatrix} = \begin{pmatrix} g(P(\sigma_0)) \\ g(P(\sigma_1)) \\ g(P(\sigma_2)) \\ \vdots \\ g(P(\sigma_N)) \end{pmatrix}$$

Remark B.1 (Analytic continuation)**.** In this step, we see that computing $f(P(\sigma_k))$ requires evaluating the real analytic function P on complex inputs. This assumes that we are able to analytically continue P to the unit disk in $\mathbb{C}$. Moreover, since the analytic continuation of P is complex valued, we have to evaluate the real analytic function g for complex values as well. Again, this may require analytic continuation of g.

Let

$$V = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^N \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^N & \omega^{2N} & \cdots & \omega^{N^2} \end{pmatrix},$$

Observe that V is invertible with

$$V^{-1} = \frac{1}{N+1} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega^{-1} & \omega^{-2} & \cdots & \omega^{-N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{-N} & \omega^{-2N} & \cdots & \omega^{-N^2} \end{pmatrix}.$$

These are the well known DFT matrices. Since we have an explicit inverse, solving the linear system determining the coefficients of the composition is reduced to matrix-vector multiplication. Moreover, when N is large, this matrix-vector multiplication is accelerated using the Fast Fourier Transform (FFT).

When P itself is a polynomial (as will be the case in our Newton procedure) we write

$$P(\sigma) = \sum_{n=0}^{N} a_n \sigma^n,$$

and have that evaluation of P at the interpolation points σ_k is expressed as

$$P(\sigma_k) = \sum_{n=0}^{N} a_n \sigma_k^n.$$

That is

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^N \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^N & \omega^{2N} & \cdots & \omega^{N^2} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix} = \begin{pmatrix} (P(q_0)) \\ (P(q_1)) \\ (P(q_2)) \\ \vdots \\ (P(q_N)) \end{pmatrix}$$

Let $p_n = P(q_k)$ for $0 \leq k \leq N$ and write

$$\mathbf{p} = \begin{pmatrix} p_0 \\ \vdots \\ p_N \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} a_0 \\ \vdots \\ a_N \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} \bar{b}_0 \\ \vdots \\ \bar{b}_N \end{pmatrix}.$$

Given the coefficients $\mathbf{a}$ of a polynomial P, the procedure for approximating the series coefficients of the composition is as follows:

- Compute $\mathbf{p}$ via
$$V\mathbf{a} = \mathbf{p}.$$
This step is actually known as the DFT.
- Compute the numbers $f_k = f(p_k)$, $0 \leq k \leq N$. Let
$$\mathbf{f} = \begin{pmatrix} \bar{f}_0 \\ \vdots \\ \bar{f}_N \end{pmatrix}.$$
- Compute
$$\mathbf{b} = V^{-1}\mathbf{f}.$$
This step is actually known as the inverse discrete Fourier transform (IDFT).

The components of $\mathbf{b}$ are the approximate Taylor coefficients of the composition $f \circ P$.

B.5. **Multiplication matrix for power series.** In the context of a Newton procedure, we often have to solve equations like

$$Q(\sigma)P(\sigma) = g(\sigma), \tag{21}$$

where Q and g are given truncated power series: that is, polynomials. (The equation is an oversimplification, but it will make the necessary point). While this equation can be solved by recursion, it is also amenable to numerical linear algebra.

To see this, let

$$\mathbf{a} = \begin{pmatrix} a_0 \\ \vdots \\ a_N \end{pmatrix}, \qquad \text{and} \qquad \mathbf{g} = \begin{pmatrix} g_0 \\ \vdots \\ g_N \end{pmatrix},$$

denote the vectors of known polynomial coefficients, and

$$\mathbf{b} = \begin{pmatrix} b_0 \\ \vdots \\ b_N \end{pmatrix},$$

denote the unknown coefficients of P (the quotient of g and Q).

Since this equation is linear in P, there is a matrix representing the action of Q on P. Applying the formula of Equation (19) to the standard basis vectors leads to a representation of the action of the Cauchy product on the coefficient vector for P given by

$$Q \cdot P = \begin{pmatrix} a_0 & 0 & 0 & \dots & 0 \\ a_1 & a_0 & 0 & \dots & 0 \\ a_2 & a_1 & a_0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_N & a_{N-1} & a_{N-2} & \vdots & a_0 \end{pmatrix} \begin{pmatrix} b_0 \\ b_1 \\ b_2 \\ \vdots \\ b_N \end{pmatrix}$$

Let

$$\text{CauchyMat}(\mathbf{a}) = \begin{pmatrix} a_0 & 0 & 0 & \dots & 0 \\ a_1 & a_0 & 0 & \dots & 0 \\ a_2 & a_1 & a_0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_N & a_{N-1} & a_{N-2} & \vdots & a_0 \end{pmatrix}. \tag{22}$$

We can solve for the unknown coefficients $\mathbf{b}$ of $P(\sigma)$ in Equation (21) by solving the (lower triangular) linear system

$$\text{CauchyMat}(\mathbf{a})\mathbf{b} = \mathbf{g}.$$

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